

ENGLISH FOR SOFTWARE ENGINEERS AND DEVELOPMENT TEAMS

Software Development English

Make the behavior clear. Make the limits explicit.

Phil Stilwell

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Eight scenarios. Eight extended conversations.

INSIDE THE BOOK

192 vocabulary entries / 128 reusable phrases

160 dialogue turns / 80 numbered dialogue gaps

Word banks and explained answer keys

Upper-intermediate to advanced (B2-C1). For software developers discussing interfaces, debugging, reviews, dependencies, tests, performance, distributed behavior, and releases.

2026 edition

Eight scenarios. Eight skills.

Eight engineering conversations follow work from an unclear interface contract to a release handoff. Practice requesting evidence, challenging assumptions, explaining tradeoffs, and distinguishing a tested result from an unverified expectation.

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Describe behavior before confidence.

A confident engineering statement names its evidence and its boundary. A test double is not the real dependency; a staging deployment is not a production release; a timeout does not establish that a request failed.

Separate contract from convention

An omitted field, an empty string, and null can mean different things. State what this interface actually promises rather than relying on how another system behaves. A proposed interpretation needs agreement before it becomes a contract.

"Omission leaves the nickname unchanged; the meaning of null is not specified."

Make disagreement actionable

Anchor review comments in a changed line, observable behavior, and a concrete request. Keep naming preferences separate from defects. A focused failure test can make the concern easier to discuss without turning the review into a judgment of its author.

"Moving this call outside the success branch also sends completion notices after failure."

Attach the measurement conditions

A median describes the middle of observed results, not the slowest users or production behavior. Compare implementations under the same stated conditions, and include resource budgets. A faster measured response does not settle an overall tradeoff.

"B lowers the median to twenty-five milliseconds but exceeds the memory budget."

Keep release stages distinct

Code can be deployed but disabled, tested in staging but not production, or approved for review but not release. Name the environment, flag state, unresolved compatibility check, and actual decision owner in the handoff.

"R18 is in staging with search disabled; Lee still owns the release decision."

PROFESSIONAL BOUNDARIES

All products, people, measurements, interfaces, and release processes are fictional. These scenarios teach professional English, not production operating instructions. Actual contracts, security requirements, engineering review, test environments, and change authorization take precedence. No exercise authorizes production changes or guarantees correctness, compatibility, reliability, or performance. Code-review and testing terminology is used without requiring a particular language or framework.

Three nickname states, only two defined meanings

A Read the situation

Distinguish omission, an empty string, and null while requesting an explicit interface decision before implementation.

A fictional profile interface accepts an optional nickname. Its current contract says an omitted nickname leaves the stored value unchanged, while an empty string clears it. A proposed client sends null. The meaning of null has not been specified. Client developer Maya and service developer Arun review the mismatch before either side implements a new interpretation. Optional describes whether the field may be absent; it does not settle what an explicit null means.

PEOPLE / Maya: Client developer | Arun: Service developer

ASK FOR THE MISSING RULE, NOT A GUESS

Developers may use the same familiar word for different behaviors. Repeat the exact input and resulting state before agreeing that an interface is clear. An explicit question about null avoids hiding a product decision inside an implementation detail.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What does omitting nickname do under the current contract?

- | | |
|-------------------------------|--|
| A Rejects every update | B Leaves the stored value unchanged |
| C Clears it | D Writes the word null |

2. What does an empty string do?

- | | |
|----------------------------------|--------------------------------------|
| A Leaves it unchanged | B Sets an unspecified default |
| C Proves null is accepted | D Clears the nickname |

3. What remains undefined?

- | | |
|--|---------------------------------------|
| A The meaning of an explicit null | B Whether omission is possible |
| C Whether an empty string clears | D Whether nickname is a field |

[Answers and explanations: page 76.](#)

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

interface contract

Agreed rules for inputs, outputs, and observable behavior.

IN USE clarify the interface contract

application programming interface

Defined way software components communicate; abbreviated API.

IN USE document the API

request payload

Data carried by a request.

IN USE inspect the request payload

optional field

Field that may be absent under the contract.

IN USE omit an optional field

nullable field

Field whose allowed values include null under the contract.

IN USE declare a nullable field

omitted field

Field not included in the submitted data.

IN USE distinguish an omitted field

empty string

Text value containing zero characters.

IN USE send an empty string

null value

Explicit marker whose meaning depends on the contract.

IN USE define the null value

stored state

Value or condition currently retained by the system.

IN USE preserve stored state

partial update

Change affecting only specified parts of a record.

IN USE apply a partial update

default value

Value supplied when the governing rules call for one.

IN USE document the default value

validation rule

Condition an input must satisfy.

IN USE agree on a validation rule

Precision check

Optional and nullable answer different questions. Optional concerns presence; nullable concerns allowed values. The stated contract defines omission and an empty string, but it does not authorize treating null as either one.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

serialization

Conversion of data into a transportable representation.

IN USE inspect serialization behavior

deserialization

Conversion from transported data into usable values.

IN USE verify deserialization behavior

schema

Formal description of data structure and constraints.

IN USE revise the schema

field presence

Whether a particular field exists in the input.

IN USE check field presence

sentinel value

Special value used to signal a particular condition.

IN USE define a sentinel value

backward compatibility

Ability to preserve supported behavior for earlier consumers.

IN USE assess backward compatibility

consumer

Component using another component's interface.

IN USE identify the consumer

producer

Component providing data or behavior through an interface.

IN USE align producer expectations

wire format

Actual representation transmitted between components.

IN USE inspect the wire format

contract test

Test checking agreed interactions across an interface boundary.

IN USE add a contract test

undefined behavior

Behavior not specified by the relevant agreement here.

IN USE resolve undefined behavior

acceptance criterion

Specific condition used to judge whether work meets requirements.

IN USE state an acceptance criterion

Precision check

Use an input-to-effect sentence: If nickname is omitted, the stored value remains unchanged. Keep the proposed null behavior explicitly unresolved until the responsible parties agree, document it, and add the appropriate checks.

Phrases you can actually use

C Notice the language / 1 of 2

NAME THE BOUNDARY

"I want to confirm the profile interface contract."

DESCRIBE CLEARING

"An empty string clears the nickname."

EXPOSE THE GAP

"The contract does not specify what null means."

ASK FOR THE EFFECT

"What should the stored value be after that input?"

DESCRIBE ABSENCE

"An omitted nickname leaves the stored value unchanged."

IDENTIFY THE PROPOSAL

"The proposed client sends null."

SEPARATE TERMS

"Optional does not automatically mean nullable."

AVOID BORROWING A RULE

"I would not infer this from another API."

Does not automatically mean

Rejects an inference without rejecting the other person's whole proposal.

After that input

Connects a request value to observable stored state.

Would be

Marks a proposed consequence rather than an existing rule.

Phrases you can actually use

C Notice the language / 2 of 2

INSPECT THE ACTUAL DATA

"Can we check the request payload?"

KEEP A PROPOSAL PROVISIONAL

"Treating null as clear would be a new decision."

REQUEST DOCUMENTATION

"Please record that rule in the contract."

AVOID PREMATURE COMPLETION

"The null case is still unresolved."

STATE A CONDITION

"If the field is absent, the current value stays."

REQUEST AGREEMENT

"Can we agree on the intended behavior first?"

REQUEST A CHECK

"We need a test for each defined input state."

CLOSE PRECISELY

"We agree on omission and empty string, but not yet on null."

Not yet

Keeps an unresolved decision visible without implying permanent disagreement.

Each defined input state

Prevents a broad test claim based on only one example.

Can we agree

Invites a concrete decision about the missing behavior.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which comment identifies the real gap?

- A** Empty string and omission are identical
- B** Every API interprets null alike
- C** The null case needs an agreed meaning
- D** Optional proves null clears it

2. Which pair must remain distinct?

- A** Null and a documented clearing command
- B** Omission and an empty string
- C** Two spellings of the same field name
- D** Two confirmed clearing commands

3. What would justify calling null a clearing command?

- A** The developer's preference alone
- B** The word optional alone
- C** An unrelated API example
- D** An explicit agreed contract change

4. Which test plan matches the discussion?

- A** Check each defined input state and resolve null before asserting its effect
- B** Test omission and assume all other inputs match
- C** Test only the client's display
- D** Treat null as proven compatible

[Answers and explanations: page 76.](#)

Three nickname states, only two defined meanings

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

interface contract / validation rule / backward compatibility / contract test / acceptance criterion / empty string / request payload / field presence / omitted field / optional field

- 01 **Maya:** Before I finish the profile client, can we check what happens when someone removes a nickname? My proposed payload sends null, but the contract mentions an empty string.
- 02 **Arun:** We should separate that from an ¹ _____. If nickname is absent from the request, the stored value stays unchanged. That part is already defined.
- 03 **Maya:** Right. I was reading optional as permission to send no value. I can see that absence and an explicit value are different on the wire.
- 04 **Arun:** Exactly. An ² _____ may be absent. It does not tell us whether null is accepted or what effect null should have if it is accepted.
- 05 **Maya:** For the existing clearing behavior, the client should send a string with no characters. That is different from leaving the property out altogether.
- 06 **Arun:** Yes, an ³ _____ clears the nickname. We should preserve that distinction in the examples so nobody silently changes the current behavior while updating the client.
- 07 **Maya:** Then my null proposal has no agreed effect yet. I should not describe it as equivalent to clearing just because another service uses it that way.
- 08 **Arun:** Correct. The ⁴ _____ needs a decision for that input. We could discuss a change, but discussion alone does not make the proposed meaning an existing guarantee.
- 09 **Maya:** Could we inspect what the client actually sends? A serializer might drop a property, and then our conversation about null would not match the transmitted data.
- 10 **Arun:** Let us check the ⁵ _____ in a controlled example. We need to distinguish the client's internal value from the representation delivered to the service.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 10.

- 11 **Maya:** I will keep three examples separate: nickname absent, nickname as an empty string, and nickname explicitly null. The third will have no claimed result until we agree.
- 12 **Arun:** That will make ⁶ _____ visible. For the first two, write the resulting stored value beside the input rather than using the vague label no nickname.
- 13 **Maya:** Should the schema be changed now to allow null? I do not want the schema to appear to settle a behavior we have not actually decided.
- 14 **Arun:** Hold that change until the ⁷ _____ and intended effect are agreed. Allowing a value and defining its effect need to be consistent.
- 15 **Maya:** Once the rule is documented, I can add client examples and ask for checks at the interface boundary, not only checks of the form display.
- 16 **Arun:** A ⁸ _____ can support that. Its expected result must come from the agreement, though; the test must not invent the missing null rule.
- 17 **Maya:** I also want existing clients protected. We should not accidentally change omission from leave unchanged to clear while solving this new input question.
- 18 **Arun:** That is a ⁹ _____ concern worth recording. We have not demonstrated compatibility merely by agreeing that the new client looks correct.
- 19 **Maya:** My summary will say omission preserves, empty string clears, and null remains unspecified. I will leave the proposed null implementation pending the interface decision.
- 20 **Arun:** Good. That gives us an explicit ¹⁰ _____ once the decision is made, instead of treating an ambiguous word as enough to approve the change.

ANSWERS / EXPLANATIONS: PAGE 77

1. omitted field | 2. optional field | 3. empty string | 4. interface contract | 5. request payload | 6. field presence | 7. validation rule | 8. contract test | 9. backward compatibility | 10. acceptance criterion

Three inputs, three separate statements

F Say it

Complete the interface summary using only the stated contract and the unresolved proposal.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

nullable / clears / unchanged / unspecified

Developer: "An omitted nickname leaves the value ¹ _____."

Reviewer: "An empty string ² _____ the nickname."

Developer: "The meaning of null remains ³ _____."

Reviewer: "Optional does not automatically mean ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 77

1. unchanged | 2. clears | 3. unspecified | 4. nullable

Rehearse with a fixed script

1. Check your answers, then read turns 1-10 with a partner using the corrected words.
2. Switch roles for turns 11-20. Stress words that distinguish facts from conditions or estimates.
3. Complete and check the four-line exchange. Read it twice, switching roles and keeping the printed facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

The empty date is a suspect, not yet the cause

A Read the situation

Report a reproducible failure, expose a confounded comparison, and request a controlled test without overstating causation.

Developer Nia has a three-row import file containing one empty date. It fails, and the stack trace points to date parsing. A two-row file without that row succeeds. Colleague Ellis reviews the evidence. No controlled test has isolated the empty date as the cause: the successful file differs in both row count and content. They need a test in a suitable nonproduction environment that changes one relevant condition while preserving the others.

PEOPLE / Nia: Developer investigating the import | Ellis: Developer reviewing the evidence

TREAT A HYPOTHESIS AS USEFUL, NOT EMBARRASSING

Naming a likely cause can help investigation, provided its status remains clear. Distinguish I suspect from I have isolated. A colleague can challenge a comparison without dismissing the observation or the person who found it.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which input fails?

- A** Every file with any date
- B** A three-row file containing an empty date
- C** Every three-row file
- D** A two-row file with all dates empty

2. What does the successful comparison change?

- A** Nothing relevant
- B** Both row count and content
- C** Only the date format
- D** Only the environment

3. What does the stack trace establish?

- A** The empty field is conclusively the root cause
- B** The database is corrupt
- C** Production is safe to test
- D** The observed failure path reaches date parsing

Answers and explanations: page 78.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

reproduction

Repeatable demonstration of an observed failure.

IN USE obtain a reproduction

reproducible example

Input and conditions that consistently show the behavior.

IN USE share a reproducible example

stack trace

Recorded chain of calls associated with an error.

IN USE inspect the stack trace

parser

Component interpreting structured input.

IN USE inspect the parser

date parsing

Conversion of date text into a usable representation.

IN USE investigate date parsing

input fixture

Prepared input used in a test.

IN USE preserve the input fixture

hypothesis

Proposed explanation requiring evidence.

IN USE test a hypothesis

root cause

Underlying cause established through investigation.

IN USE isolate the root cause

confounding variable

Additional changed factor obscuring a comparison.

IN USE remove a confounding variable

controlled comparison

Comparison holding relevant conditions fixed except the tested change.

IN USE run a controlled comparison

minimal example

Reduced example retaining the relevant behavior.

IN USE reduce to a minimal example

baseline

Reference condition against which changes are compared.

IN USE establish a baseline

Precision check

The two-row success is useful evidence, but it changes two things at once. It does not distinguish an empty-date effect from other differences associated with the removed row or the changed row count.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

failure signature

Identifying pattern of an observed failure.

IN USE compare the failure signature

exception

Signal that interrupts normal execution under specified conditions.

IN USE capture the exception

log context

Surrounding recorded information explaining an event.

IN USE preserve log context

row count

Number of records or lines in the input table.

IN USE hold row count constant

empty field

Present field containing no supplied value.

IN USE isolate the empty field

malformed input

Data not conforming to the required structure or rules.

IN USE identify malformed input

deterministic behavior

Behavior producing the same result under the same relevant conditions.

IN USE check deterministic behavior

environment parity

Relevant similarity between execution environments.

IN USE establish environment parity

regression

Previously working behavior broken by a later change.

IN USE investigate a regression

instrumentation

Added observation mechanisms such as logs or measurements.

IN USE improve instrumentation

causal claim

Statement that one condition produces an effect.

IN USE qualify a causal claim

nonproduction environment

Environment not serving the live production workload.

IN USE use a nonproduction environment

Precision check

A stack trace helps locate where the failure surfaced. It is not automatically a root-cause analysis. Propose a controlled comparison with the same row count and other content while varying the suspect date condition.

Phrases you can actually use

C Notice the language / 1 of 2

REPORT THE FAILING INPUT

"The three-row file fails during import."

NAME THE OBSERVATION

"The stack trace reaches date parsing."

QUALIFY THE CONCLUSION

"I suspect the empty date, but have not isolated it."

ASK FOR CONTROL

"Can we hold the other fields constant?"

REPORT THE DISTINGUISHING DETAIL

"One row contains an empty date."

REPORT THE COMPARISON

"The two-row file succeeds."

EXPOSE THE CONFOUND

"We changed both row count and content."

PROPOSE A NARROW CHANGE

"Change only the suspect date condition."

I suspect

Gives a working explanation an explicitly provisional status.

Have not isolated

Identifies the missing causal evidence.

Both ... and

Makes the two changing factors visible.

Phrases you can actually use

C Notice the language / 2 of 2

PRESERVE EVIDENCE

"Keep the original failing fixture."

ASK FOR REPEATABILITY

"Does the same fixture fail consistently?"

NAME THE SETTING

"Run the comparison in a suitable nonproduction environment."

REPORT A LIMIT

"We do not have the controlled result yet."

AVOID A UNIVERSAL CLAIM

"We have not shown that every empty date fails."

COMPARE THE OUTCOME

"Check whether the failure signature stays the same."

SEPARATE LOCATION AND CAUSE

"The trace points to parsing, not yet to a proven cause."

CLOSE THE HANDOFF

"Share the fixture, conditions, trace, and next comparison."

Hold constant

Requests an unchanged comparison condition.

Not yet

Limits the conclusion without discarding the evidence.

Whether

Introduces a check whose result is genuinely unresolved.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which conclusion is supported?

- A** The root cause is proven
- B** The observed failing path reaches date parsing
- C** All empty dates always fail
- D** Row count has no effect

2. Which proposed comparison is strongest?

- A** Change date, encoding, and parser version together
- B** Keep three rows and other content fixed while varying the suspect date
- C** Delete two more rows and change the environment
- D** Compare unrelated files

3. Which phrase avoids overstating evidence?

- A** The empty date is conclusively responsible
- B** The trace proves every input rule
- C** The successful file eliminates all other causes
- D** The empty date is a working hypothesis

4. What should the handoff preserve?

- A** The failing fixture, conditions, and trace
- B** Only the developer's confidence
- C** A rewritten file without its original data
- D** A production change without authorization

Answers and explanations: [page 78](#).

The empty date is a suspect, not yet the cause

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

deterministic behavior / reproducible example / hypothesis / confounding variable /
nonproduction environment / stack trace / failure signature / input fixture / causal claim /
controlled comparison

- 01 **Nia:** I can reproduce the import failure with this three-row file. One row has an empty date, and the trace ends in the date-parsing path.
- 02 **Ellis:** That gives us a useful ¹ _____. What did you change in the file that succeeds, and did you keep the other conditions the same?
- 03 **Nia:** I removed the row with the empty date. The remaining two rows import successfully. I was about to report that the empty date causes the failure.
- 04 **Ellis:** I would call that a ² _____ for now. Removing the row changed its content and the total number of rows, so the comparison does not isolate the date.
- 05 **Nia:** Fair point. The successful file tells us the smaller input works, but it cannot show which part of the removed row matters.
- 06 **Ellis:** Exactly. The changed row count is a possible ³ _____. We should avoid ruling it out before making a comparison that keeps the relevant conditions fixed.
- 07 **Nia:** Would keeping all three rows and changing only that date give us a clearer test? I would preserve the other fields and use the same environment.
- 08 **Ellis:** That is a ⁴ _____ worth preparing. We still need to run it and record the outcome; a proposed test is not evidence of its result.
- 09 **Nia:** I will keep the original file untouched as a reference. The comparison file can be separate, with the specific change recorded beside it.
- 10 **Ellis:** Good. Preserve the failing ⁵ _____ so another developer can reproduce your starting point. Otherwise we may accidentally lose the very example that made the problem observable.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 19.

- 11 **Nia:** How should I describe the trace in the issue? I do not want to make it sound irrelevant just because it does not settle causation.
- 12 **Ellis:** Say the ⁶ _____ points to date parsing. That is useful location evidence. Keep it separate from the stronger claim that a particular input condition caused the failure.
- 13 **Nia:** I should also record the exact error pattern. If the modified file fails differently, saying both failed could hide an important distinction.
- 14 **Ellis:** Yes, compare the ⁷ _____, not just the final status. A different exception or call path may mean you are looking at another problem rather than reproducing the same one.
- 15 **Nia:** I have not checked repeated runs yet. The example failed when I tried it, but I should confirm how consistently it behaves under those conditions.
- 16 **Ellis:** That helps establish ⁸ _____. Record what you actually observe rather than assuming repeatability from one successful reproduction.
- 17 **Nia:** This should stay outside production. We can prepare the comparison in an appropriate test setting without asking live users to trigger the error.
- 18 **Ellis:** Agreed. Use a suitable ⁹ _____ and record its relevant conditions. A result there still needs its environment boundary stated when you pass it on.
- 19 **Nia:** I will update the issue to say the empty date is suspected, the two-row comparison is inconclusive, and the controlled three-row test is pending.
- 20 **Ellis:** That is a well-qualified ¹⁰ _____. Include the fixture and trace, and we will assess the controlled result before calling anything the root cause.

ANSWERS / EXPLANATIONS: PAGE 79

1. reproducible example | 2. hypothesis | 3. confounding variable | 4. controlled comparison | 5. input fixture | 6. stack trace | 7. failure signature | 8. deterministic behavior | 9. nonproduction environment | 10. causal claim

Report the evidence without upgrading the conclusion

F Say it

Complete the debugging handoff. The controlled test has not yet produced a result.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

two / hypothesis / three / constant

Developer: "The failing file contains ¹ _____ rows."

Reviewer: "The successful file contains ² _____ rows."

Developer: "The empty date remains a ³ _____."

Reviewer: "Keep the other conditions ⁴ _____ in the next comparison."

ANSWERS / EXPLANATIONS: PAGE 79

1. three | 2. two | 3. hypothesis | 4. constant

Rehearse with a fixed script

1. Check your answers, then read turns 1-10 with a partner using the corrected words.
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3. Complete and check the four-line exchange. Read it twice, switching roles and keeping the printed facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

A rename that changed the failure path

A Read the situation

Anchor a review comment in a changed call location, its observable effect, and a specific correction and test.

A pull request is titled Rename export helper. The diff also moves the completion-notification call outside the success branch. Failed exports therefore notify users as though export completed. Reviewer Priya accepts the naming change but raises the behavior change with author Tomas. The review should request restoration of success-only notification and a failure-path test. The acceptable rename does not justify approving the unintended notification behavior.

PEOPLE / Priya: Code reviewer | Tomas: Pull-request author

CRITIQUE THE CHANGE, NOT THE AUTHOR

A useful review points to behavior and asks for a verifiable change. Acknowledge acceptable parts without letting praise obscure a defect. Separating naming from functionality lets the author respond to a concrete concern instead of defending personal competence.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What is the pull request titled?

- A** Release the importer
- B** Rename export helper
- C** Rewrite notification policy
- D** Remove all export tests

2. What additional change appears in the diff?

- A** Notifications are removed entirely
- B** Notification moves outside the success branch
- C** The export always succeeds
- D** The failure branch is proven unreachable

3. What part does Priya accept?

- A** The false success message
- B** Missing failure coverage
- C** Automatic production release
- D** The naming change

Answers and explanations: [page 80](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

pull request

Proposed code change submitted for review; abbreviated PR.

IN USE review the pull request

diff

Display of changes between versions.

IN USE inspect the diff

review comment

Feedback attached to a proposed change.

IN USE write a review comment

success branch

Execution path entered when the relevant operation succeeds.

IN USE keep the call in the success branch

failure path

Execution path taken when an operation fails.

IN USE cover the failure path

call site

Location where a function or operation is invoked.

IN USE identify the call site

control flow

Order and conditions under which code executes.

IN USE trace the control flow

side effect

Observable effect beyond a function's returned value.

IN USE examine the side effect

completion notification

Message indicating an operation has finished successfully here.

IN USE gate the completion notification

behavior change

Alteration in what the system observably does.

IN USE flag the behavior change

refactor

Internal restructuring intended to preserve observable behavior.

IN USE separate a refactor from a behavior change

rename

Change to an identifier's name.

IN USE approve the rename

Precision check

A rename can be acceptable while the same diff contains a blocking defect. Moving the notification outside the success branch changes observable behavior, so it is not merely a naming preference or a behavior-preserving refactor.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

regression test

Test guarding against recurrence of a known defect.

IN USE add a regression test

assertion

Test statement checking an expected condition.

IN USE strengthen the assertion

blocking concern

Issue that must be resolved before approval under the review process.

IN USE explain the blocking concern

nonblocking suggestion

Improvement that does not prevent approval under the process.

IN USE label a nonblocking suggestion

review scope

Set of changes and concerns covered by a review.

IN USE clarify review scope

intent

Purpose the author means a change to achieve.

IN USE confirm the intent

observable behavior

Effect visible to a user or another component.

IN USE preserve observable behavior

false success

Indication of success when the operation did not succeed.

IN USE prevent false success

guard condition

Condition controlling whether an action may occur.

IN USE restore the guard condition

patch revision

Updated version of a proposed change.

IN USE request a patch revision

approval

Explicit acceptance at the relevant review stage.

IN USE withhold approval pending correction

verification evidence

Results supporting a claim that a change behaves as intended.

IN USE provide verification evidence

Precision check

Make the request testable: restore success-only notification, then add a check that a failed export does not produce a completion notice. Do not substitute a general request to improve code quality for the specific observed issue.

Phrases you can actually use

C Notice the language / 1 of 2

03

ACKNOWLEDGE THE ACCEPTABLE PART

"The helper name is clear."

DESCRIBE THE CONSEQUENCE

"A failed export can now send a completion notice."

ASK ABOUT INTENT

"Was this control-flow change intentional?"

MAKE A CONCRETE REQUEST

"Please restore the success-only condition."

LOCATE THE CONCERN

"This notification call has moved outside the success branch."

SEPARATE PREFERENCE AND DEFECT

"My concern is behavior, not the naming choice."

STATE THE EXPECTED BOUNDARY

"The completion notice belongs on the success path."

REQUEST EVIDENCE

"Add a test for a failed export."

This call

Anchors the concern to a particular location in the change.

Can now

Identifies a newly possible behavior rather than a style preference.

Was ... intentional

Asks about purpose without assuming carelessness.

Phrases you can actually use

C Notice the language / 2 of 2

SPECIFY THE ASSERTION

"The failure case should not send a completion notification."

CLARIFY REVIEW STATUS

"I cannot approve this behavior yet."

DISTINGUISH SCOPE

"The diff does more than the title suggests."

KEEP VERIFICATION SEPARATE

"I will review the revised behavior and test."

AVOID PERSONAL CRITICISM

"This change introduces a false-success path."

LIMIT THE OBJECTION

"I am not asking you to undo the rename."

REQUEST A REVISION

"Please update the patch and its description."

CLOSE CONSTRUCTIVELY

"The rename can stay once the notification issue is resolved."

Not ... but

Separates the accepted naming issue from the behavioral objection.

Should not

States the negative assertion needed in the failure test.

Once

Makes approval conditional on a specific unresolved correction.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which comment is actionable?

- A** Restore the success-only notification and test that failure sends no completion notice
- B** Your code is careless
- C** Make this nicer
- D** The name is fine, so everything is approved

2. Why is this not only a refactor?

- A** Observable notification behavior changes
- B** The helper has a new name
- C** Review comments are present
- D** The title is short

3. What should the failure test assert?

- A** Production release is approved
- B** No completion notice is sent
- C** A completion notice always appears
- D** Only the helper spelling is correct

4. Which statement preserves Priya's position?

- A** The rename is acceptable; the notification behavior needs correction
- B** Every line must be reverted
- C** The author cannot write software
- D** The test can be omitted because the rename is small

Answers and explanations: [page 80](#).

A rename that changed the failure path

E Conversations / turns 1-10

03

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

verification evidence / side effect / patch revision / control flow / rename / false success / failure path / review comment / regression test / review scope

- 01 **Priya:** The new helper name makes sense to me. I have one blocking concern in the same diff: the completion-notification call has moved outside the success branch.
- 02 **Tomas:** I intended this as a ¹ _____, so I was not expecting a behavior objection. Can you point me to the consequence you are seeing rather than the name itself?
- 03 **Priya:** When export fails, execution can still reach that notification call. Users then receive a completion message even though the export did not complete successfully.
- 04 **Tomas:** I see the changed ² _____ now. The call used to depend on success, and the new position removes that condition. That was not my intended change.
- 05 **Priya:** Thanks. I am not asking you to undo the clearer name. I am asking for the notification to remain conditional on a successful export.
- 06 **Tomas:** So the concern is the ³ _____, not the naming style. I can keep the rename and restore the original success-only placement.
- 07 **Priya:** Yes. Please also add a test where export fails and verify that no completion notification is sent. The success case alone will not cover this problem.
- 08 **Tomas:** That gives the ⁴ _____ an explicit expectation. I will not just assert that the helper was called; I need to check the notification does not happen.
- 09 **Priya:** Exactly. The review comment will identify the moved call, explain the false completion message, and request the condition and test. That should make the revision clear.
- 10 **Tomas:** I appreciate the specific ⁵ _____. A general request to improve the code would not tell me which behavior you need preserved.

The conversation, continued

E Conversations / turns 11-20

03

Continue with the word bank on page 28.

- 11 **Priya:** The title also understates the current diff. Once you revise it, please make sure the description accurately reflects what remains and why the test was added.
- 12 **Tomas:** I will update the ⁶ _____ rather than assume the title proves this is harmless. Reviewers still need to examine the changed lines.
- 13 **Priya:** Calling the whole change a refactor would be misleading in its present form. A behavior-preserving change should not turn failed exports into apparent successes.
- 14 **Tomas:** Agreed. The ⁷ _____ is the defect here. It is separate from whether the new name is clearer or the patch is small.
- 15 **Priya:** After your revision, I will check the call placement and the failure assertion. Please include the test result, but do not describe my current feedback as approval.
- 16 **Tomas:** Understood. I will submit a ⁸ _____ with the rename intact, the notification guarded by success, and evidence from the added test.
- 17 **Priya:** The test should be specific enough to fail if someone moves the call out again. Otherwise it could pass while the same user-visible problem returns.
- 18 **Tomas:** That makes it a useful ⁹ _____. I will check the negative notification expectation, not only that the test reaches the export code.
- 19 **Priya:** Good. My final comment will say the naming change is acceptable and the notification issue blocks approval until corrected and verified. That keeps the two points distinct.
- 20 **Tomas:** Then the next step is ¹⁰ _____, not a debate about naming. I will revise the patch and return it for review with the relevant result.

ANSWERS / EXPLANATIONS: PAGE 81

1. rename | 2. control flow | 3. side effect | 4. failure path | 5. review comment | 6. review scope | 7. false success | 8. patch revision | 9. regression test | 10. verification evidence

Turn a vague objection into a review request

F Say it

Complete the review summary while preserving the accepted rename and unresolved behavior issue.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

failure / approval / success / name

Reviewer: "The helper ¹ _____ is acceptable."

Author: "The notification must remain on the ² _____ path."

Reviewer: "Add a test for the ³ _____ path."

Author: "The current feedback is not yet ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 81

1. name | 2. success | 3. failure | 4. approval

Rehearse with a fixed script

1. Check your answers, then read turns 1-10 with a partner using the corrected words.
2. Switch roles for turns 11-20. Stress words that distinguish facts from conditions or estimates.
3. Complete and check the four-line exchange. Read it twice, switching roles and keeping the printed facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

A direct upgrade with an indirect change

04

A Read the situation

Request the resolved dependency version, affected call sites, and migration evidence before claiming upgrade compatibility.

A fictional application uses ParcelKit directly. ParcelKit in turn uses TextCore. The proposed ParcelKit upgrade changes the resolved TextCore version. A change note marks a parsing entry point as deprecated. Developers Leila and Ben have not established whether the application or its dependencies use that entry point, whether it has been removed, or whether the replacement is compatible. The packages are not assumed to follow any particular versioning convention.

PEOPLE / Leila: Developer proposing the upgrade | Ben: Developer checking dependency impact

ASK WHAT CHANGED BELOW THE HEADLINE

An upgrade can affect more than the package named in the request. Asking for the resolved dependency tree and affected calls is a practical evidence request, not resistance to maintenance. Avoid translating deprecated into removed or compatible without checking the actual contract.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which dependency does the application use directly?

A Neither

B ParcelKit

C TextCore only

D Both are proven direct

2. What changes indirectly?

A Every public interface

B A confirmed production failure

C The resolved TextCore version

D The application name

3. What does the note establish?

A A parsing entry point is deprecated

B The replacement is fully compatible

C The application definitely uses it

D The entry point has definitely been removed

Answers and explanations: page 82.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

direct dependency

Package explicitly used or declared by the application here.

IN USE upgrade a direct dependency

transitive dependency

Package brought in through another dependency.

IN USE inspect a transitive dependency

dependency tree

Structure showing how packages depend on one another.

IN USE inspect the dependency tree

resolved version

Specific version selected after dependency resolution.

IN USE record the resolved version

lockfile

File recording resolved dependency selections for reproducibility.

IN USE inspect the lockfile

version constraint

Rule limiting which dependency versions may be selected.

IN USE check a version constraint

release note

Published description of changes in a release.

IN USE read the release note

deprecation

Notice that an interface is discouraged or scheduled for replacement.

IN USE investigate a deprecation

removal

Elimination of a previously available interface or behavior.

IN USE verify removal separately

migration guide

Instructions describing transition to a changed interface.

IN USE consult the migration guide

entry point

Named interface through which functionality is invoked.

IN USE locate the parsing entry point

call site

Location that invokes a particular interface.

IN USE search for affected call sites

Precision check

Direct describes the application-to-ParcelKit relationship. Transitive describes the application-to-TextCore relationship through ParcelKit. The dependency actually selected matters, not just the version written in the top-level upgrade request.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

public interface

Supported surface exposed for other components to use.

IN USE compare the public interface

breaking change

Change that can invalidate previously supported use.

IN USE identify a breaking change

compatibility

Ability to work with the relevant existing consumers and behavior.

IN USE verify compatibility

semantic versioning

Version convention relating version numbers to public-interface changes.

IN USE confirm semantic versioning policy

major version

Leading version component under a stated convention.

IN USE interpret the major version

minor version

Version component whose meaning depends on the adopted convention.

IN USE check minor-version guarantees

patch version

Version component commonly used for fixes under a stated convention.

IN USE verify patch-version policy

pin

Fixed dependency selection restricting version movement.

IN USE review the version pin

changelog

Maintained record of notable changes between versions.

IN USE compare the changelog

replacement interface

Interface intended to supersede another.

IN USE assess the replacement interface

resolution

Process selecting concrete dependency versions.

IN USE reproduce dependency resolution

upgrade evidence

Records and checks supporting an upgrade assessment.

IN USE assemble upgrade evidence

Precision check

Deprecated does not itself mean removed, unused, or compatible with a replacement. Version numbers provide guarantees only under the package policy that actually applies. Request usage and migration evidence before declaring the upgrade safe.

Phrases you can actually use

C Notice the language / 1 of 2

IDENTIFY THE DIRECT DEPENDENCY

"The application uses ParcelKit directly."

NAME THE ADDITIONAL CHANGE

"The resolved TextCore version also changes."

ASK FOR THE RELATIONSHIP

"Can you show the dependency tree?"

AVOID AN UNSUPPORTED UPGRADE

"Deprecated does not by itself mean removed."

IDENTIFY THE INDIRECT DEPENDENCY

"TextCore comes through ParcelKit."

REQUEST THE ACTUAL SELECTION

"Which version does the lockfile record?"

QUOTE THE NARROW FINDING

"The note marks the parsing entry point as deprecated."

ASK ABOUT USAGE

"Do we have any affected call sites?"

Comes through

Expresses an indirect dependency relationship in ordinary speech.

Also changes

Exposes a second effect hidden by the top-level upgrade description.

By itself

Limits what the deprecation label establishes.

Phrases you can actually use

C Notice the language / 2 of 2

INCLUDE INDIRECT CALLERS

"Check whether ParcelKit invokes it as well."

SEPARATE AVAILABILITY AND BEHAVIOR

"An available replacement is not automatically compatible."

REQUEST VERIFICATION

"Which checks exercise the affected parsing behavior?"

STATE THE PENDING WORK

"Usage and replacement behavior still need review."

REQUEST MIGRATION DETAIL

"What does the migration guide say about the replacement?"

QUALIFY THE VERSION INFERENCE

"We need the package's actual versioning policy."

KEEP THE DECISION OPEN

"We do not have enough evidence to claim compatibility."

CLOSE THE REQUEST

"Please attach the resolved versions and the relevant findings."

Whether

Leaves actual usage unresolved.

Not automatically

Challenges a shortcut inference without claiming the opposite.

Actual

Directs attention to the selected versions and applicable policy.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which statement is accurate?

- A** TextCore is transitive through ParcelKit
- B** TextCore cannot affect the application
- C** ParcelKit is only transitive
- D** All transitive dependencies are unused

2. What should be checked first about the selected version?

- A** The latest version mentioned in a blog
- B** The resolved dependency record
- C** Only the pull-request title
- D** An unrelated package's policy

3. What does deprecated not establish?

- A** That the interface deserves attention
- B** That the entry point has been removed
- C** That a transition may need review
- D** That a note exists

4. Which compatibility claim is justified now?

- A** The replacement is guaranteed identical
- B** Every minor upgrade is safe
- C** No call-site review is needed
- D** Compatibility remains unverified

Answers and explanations: [page 82](#).

A direct upgrade with an indirect change

E Conversations / turns 1-10

04

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

call site / deprecation / semantic versioning / compatibility / migration guide / upgrade evidence / dependency tree / transitive dependency / public interface / lockfile

- 01 **Leila:** I have proposed the ParcelKit upgrade. The headline change looks small, but the resolved dependency list also shows a different TextCore version.
- 02 **Ben:** Then we need to review the ¹ _____ change as well. The application uses ParcelKit directly, but that does not make TextCore irrelevant to its behavior.
- 03 **Leila:** I can attach the resolved versions before and after. Would the package declaration alone be enough, or do you want the selection recorded by the build?
- 04 **Ben:** Please include the ² _____ and the relevant dependency relationship. A constraint can allow several versions, while the resolved record tells us which one this change actually selects.
- 05 **Leila:** There is also a note about a deprecated parsing entry point. I have not found whether we call it or whether ParcelKit calls it internally.
- 06 **Ben:** Keep that as a ³ _____ finding, not a removal finding. The note deserves investigation, but its label alone does not tell us whether this application is affected.
- 07 **Leila:** So I should not write that the upgrade removes a parser we use. Both removal and our usage are still unverified.
- 08 **Ben:** Correct. Start by locating any relevant ⁴ _____, including indirect calls through ParcelKit. An absence of direct application calls would not automatically rule out an indirect dependency on that behavior.
- 09 **Leila:** If there is a replacement entry point, could I simply switch to it? The name suggests it does the same kind of parsing.
- 10 **Ben:** Consult the ⁵ _____ first. Similar names do not establish identical inputs, outputs, error behavior, or handling of edge cases. We need the relevant documented differences.

The conversation, continued

E Conversations / turns 11-20

04

Continue with the word bank on page 37.

- 11 **Leila:** I was also tempted to rely on the version number. The upgrade request does not tell us what versioning policy these packages follow.
- 12 **Ben:** Then do not assume ⁶ _____ guarantees. We can check the actual policy, but a familiar number pattern is not evidence that its promised compatibility rules apply.
- 13 **Leila:** I will separate the questions: selected versions, affected usage, the old interface status, and the replacement's behavior. That should keep the review focused.
- 14 **Ben:** Good. The ⁷ _____ will help explain why TextCore belongs in the discussion even though the request is named after ParcelKit.
- 15 **Leila:** What should the test evidence cover once we identify the affected calls? Running the existing suite is useful, but I do not know whether it reaches them.
- 16 **Ben:** Ask which checks exercise the relevant ⁸ _____. A passing suite cannot establish coverage of a path it never executes.
- 17 **Leila:** That makes sense. I can list any coverage gap explicitly instead of using all tests passed as a blanket compatibility claim.
- 18 **Ben:** Exactly. We need ⁹ _____ tied to this change. It should show what was checked and what remains unknown, not just that an upgrade command succeeded.
- 19 **Leila:** My revised summary will say ParcelKit changes the resolved TextCore version, the parsing entry point is deprecated, and usage and replacement behavior still need review.
- 20 **Ben:** That is accurate. Keep ¹⁰ _____ unverified for now. Once those findings are available, we can discuss the upgrade on evidence rather than infer its impact from the headline.

ANSWERS / EXPLANATIONS: PAGE 83

1. transitive dependency | 2. lockfile | 3. deprecation | 4. call site | 5. migration guide | 6. semantic versioning | 7. dependency tree | 8. public interface | 9. upgrade evidence | 10. compatibility

Keep the dependency chain and uncertainty intact

F Say it

Complete the upgrade handoff without turning deprecation into confirmed removal or compatibility.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

deprecated / unverified / transitive / direct

Developer: "ParcelKit is the ¹ _____ dependency."

Reviewer: "TextCore is the ² _____ dependency."

Developer: "The parsing entry point is marked ³ _____."

Reviewer: "Replacement compatibility remains ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 83

1. direct | 2. transitive | 3. deprecated | 4. unverified

Rehearse with a fixed script

1. Check your answers, then read turns 1-10 with a partner using the corrected words.
2. Switch roles for turns 11-20. Stress words that distinguish facts from conditions or estimates.
3. Complete and check the four-line exchange. Read it twice, switching roles and keeping the printed facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

A green test that never sees rejection

A Read the situation

Explain the limits of a success-only test double and distinguish a failure-path unit test from real-adapter integration coverage.

05

A save-operation unit test uses a storage double that always succeeds and checks the success message. A previous bug showed success after storage rejected the save. The current test never produces rejection, and the real storage adapter is untested. Developers Owen and Farah need to configure a rejection case, assert that false success does not appear, and keep real-adapter integration coverage as a separate unresolved concern.

PEOPLE / Owen: Developer maintaining the test | Farah: Developer reviewing coverage

A PASSING RESULT NEEDS A SCOPE STATEMENT

Green is a useful status, but it is not a complete explanation. Name the behavior the test actually exercises and the dependencies it replaces. This allows a colleague to request missing coverage without denying the value of existing tests.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What does the storage double currently do?

- A Always succeeds
- C Uses the real adapter

- B Rejects every request
- D Alternates unpredictably

2. What did the previous bug show?

- A A missing helper name
- C Success after storage rejected the save

- B A proven adapter timeout
- D Failure after a confirmed save

3. What remains separately untested?

- A The existence of the double
- C The real storage adapter

- B Every application feature
- D Whether the test shows a success message

Answers and explanations: [page 84](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

unit test

Focused test of a small behavior or component.

IN USE extend the unit test

test double

Substitute for a real dependency during testing.

IN USE configure the test double

stub

Test substitute supplying predetermined responses.

IN USE use a success stub

mock

Test substitute that can record or check interactions.

IN USE inspect mock interactions

fixture

Prepared state or data used by a test.

IN USE arrange the fixture

success response

Result indicating an operation succeeded.

IN USE return a success response

rejection

Failed outcome reported by the storage operation here.

IN USE simulate a rejection

exception

Error signal raised during execution under the relevant model.

IN USE configure an exception

assertion

Explicit check of an expected test condition.

IN USE add a negative assertion

negative assertion

Check that an unwanted event or state does not occur.

IN USE verify a negative assertion

test isolation

Separation of a tested unit from unrelated dependencies.

IN USE preserve test isolation

integration test

Test of interaction between connected components.

IN USE add an integration test

Precision check

A double that always succeeds can support a success-path assertion, but it cannot exercise rejection handling. Configure a failing response and check the user-visible consequence relevant to the known bug.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

adapter

Component translating between an application and an external interface.

IN USE test the storage adapter

branch coverage

Measure of which decision outcomes a test set executes.

IN USE examine branch coverage

happy path

Expected successful execution path.

IN USE cover the happy path

error path

Execution path handling an unsuccessful operation.

IN USE exercise the error path

false negative

Test result that misses a defect which is present.

IN USE avoid a false negative

test oracle

Rule or reference used to judge the expected result.

IN USE define the test oracle

arrange

Prepare the conditions for a test.

IN USE arrange a rejection case

act

Execute the behavior being tested.

IN USE act on the save request

assert

Check the observed result against the expectation.

IN USE assert no false success

interaction verification

Check of how a component called its dependencies.

IN USE separate interaction verification

behavioral verification

Check of the externally observable result.

IN USE perform behavioral verification

coverage gap

Relevant behavior not exercised by the current tests.

IN USE report the coverage gap

Precision check

The rejection unit test can verify application behavior against a simulated storage outcome. It does not establish that the real adapter produces that outcome correctly or that the actual integration behaves the same way.

Phrases you can actually use

C Notice the language / 1 of 2

DESCRIBE THE CURRENT SETUP

"The storage double always succeeds."

RECALL THE DEFECT

"The bug showed success after a rejected save."

REQUEST A CONFIGURED OUTCOME

"Make the double reject in a separate case."

PRESERVE EXISTING COVERAGE

"Keep the successful-save case too."

NAME THE TESTED PATH

"This test exercises the happy path."

IDENTIFY THE GAP

"The current test never triggers rejection."

SPECIFY THE CHECK

"Assert that no success message appears after rejection."

SEPARATE CALL AND RESULT

"A storage call alone does not prove the right user message."

Always

Exposes a fixture that cannot produce the missing failure condition.

After rejection

Connects the assertion to the relevant event sequence.

Separate case

Preserves success coverage while adding failure coverage.

Phrases you can actually use

C Notice the language / 2 of 2

NAME THE SUBSTITUTE

"This is a double, not the real adapter."

REQUEST ANOTHER TEST LEVEL

"The real adapter needs separate integration coverage."

STATE THE OPEN QUESTION

"We have not verified the real adapter."

REQUEST EVIDENCE

"Please include the configured outcome and assertion."

LIMIT THE CONCLUSION

"The result applies to the simulated rejection case."

AVOID A BLANKET CLAIM

"Green tests do not establish all storage behavior."

ASK ABOUT EXPECTATIONS

"What observable result should this case check?"

CLOSE ACCURATELY

"The new unit case can close one gap, not every storage risk."

Alone

Limits what a dependency-call assertion can demonstrate.

Simulated

Marks the boundary between a double and real integration.

One gap

Avoids overstating the scope of the proposed improvement.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Why does the current test miss the prior bug?

- A** Every unit test uses production storage
- B** Success messages cannot be tested
- C** Its double never rejects
- D** Its name contains save

2. Which addition targets the defect?

- A** Check only that storage is called
- B** Remove all assertions
- C** Configure rejection and assert no success message
- D** Rename the success test only

3. What does the rejection unit test not establish?

- A** Correct behavior of the real storage adapter
- B** Behavior under the configured double response
- C** Whether the assertion passes in that case
- D** Whether the fixture rejects

4. Which report is precise?

- A** Rejection cannot be simulated
- B** Success-path coverage exists; rejection and real-adapter coverage need work
- C** All storage behavior is verified
- D** A green test proves no bugs remain

Answers and explanations: [page 84](#).

A green test that never sees rejection

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

05

WORD + PHRASE BANK

Interaction verification / behavioral verification / arrange / integration test / happy path / success response / negative assertion / test double / unit test / coverage gap

- 01 **Owen:** The save test is green, but I am not sure it would catch the old bug. It checks the success message after the storage double returns success.
- 02 **Farah:** Then it exercises the ¹ _____. What happens in the test when storage rejects the save, which was the condition behind the earlier false success message?
- 03 **Owen:** Nothing produces rejection in the current fixture. The double always succeeds, so that condition is never reached. I should not claim the old defect is covered.
- 04 **Farah:** Right. The ² _____ is useful, but its configured behavior limits the test. We need a separate case that makes storage report the relevant unsuccessful outcome.
- 05 **Owen:** I can add that case while preserving the successful-save test. Then the new check should concern the displayed result, not just whether storage was called.
- 06 **Farah:** Add a ³ _____ that no success message appears after rejection. Calling storage is compatible with both correct handling and the old bug.
- 07 **Owen:** That distinction helps. A call-count check could pass even if the application ignored the rejected result and displayed completion anyway.
- 08 **Farah:** Exactly. ⁴ _____ is not the same as verifying the resulting message. The assertion needs to match the defect we are trying to prevent.
- 09 **Owen:** Should I describe the new case as an integration test? It crosses from the save operation to the storage interface, but the real adapter is still replaced.
- 10 **Farah:** No. The relevant ⁵ _____ would need to exercise the actual components whose interaction you are claiming to verify. This proposed case still uses a substitute.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 46.

- 11 **Owen:** Then I will state that it checks application behavior under a simulated rejection. It does not establish how the real adapter responds.
- 12 **Farah:** That keeps the ⁶ _____ visible. We know the real adapter is untested, and changing the double does not make that separate fact disappear.
- 13 **Owen:** For the new case, I will prepare the rejection response, execute the save, and inspect the resulting user message. That gives the test a clear sequence.
- 14 **Farah:** Yes: ⁷ _____ the condition, act on the save request, and assert the expected result. Keep the failure setup explicit so reviewers can see what the test really exercises.
- 15 **Owen:** The success-only test was not worthless, then. Its result was simply being described too broadly as proof of the whole storage workflow.
- 16 **Farah:** Correct. It provides evidence for its ⁸ _____ case. The problem is the wider claim, not the existence of a focused test for a successful save.
- 17 **Owen:** When I report the new result, I will name the double's configured outcome and the no-success assertion. I will list actual adapter behavior separately as unverified.
- 18 **Farah:** Good. That is ⁹ _____ within a clear boundary. Avoid saying storage is fully tested when the evidence concerns only these simulated responses.
- 19 **Owen:** The handoff will retain the happy-path test, add the rejection case, and call out the missing real-adapter check. Approval still depends on reviewing the actual revision.
- 20 **Farah:** Exactly. A targeted ¹⁰ _____ closes a specific omission. We should welcome that improvement while remaining precise about what it does and does not demonstrate.

ANSWERS / EXPLANATIONS: PAGE 85

1. happy path | 2. test double | 3. negative assertion | 4. Interaction verification | 5. integration test | 6. coverage gap | 7. arrange | 8. success response | 9. behavioral verification | 10. unit test

Name the test boundary

F Say it

Complete the review note about the current test and the proposed rejection case.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

rejection / untested / success / succeeds

Developer: "The existing double always ¹ _____."

Reviewer: "The new case must trigger ² _____."

Developer: "The assertion should prevent a false ³ _____ message."

Reviewer: "The real adapter remains ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 85

1. succeeds | 2. rejection | 3. success | 4. untested

Rehearse with a fixed script

1. Check your answers, then read turns 1-10 with a partner using the corrected words.
2. Switch roles for turns 11-20. Stress words that distinguish facts from conditions or estimates.
3. Complete and check the four-line exchange. Read it twice, switching roles and keeping the printed facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Fifteen milliseconds faster, forty megabytes over budget

A Read the situation

Compare measured latency and memory under matched conditions while separating median improvement from budget compliance and production claims.

06

In a controlled benchmark using the same input and machine, implementation A has median latency of 40 milliseconds and memory use of 80 megabytes. B has median latency of 25 milliseconds and memory use of 160 megabytes. The memory budget is 120 megabytes. Developers Sora and Malik discuss the tradeoff. Tail latency and production behavior have not been measured. B reduces the measured median by 15 milliseconds, or 37.5 percent relative to A, but exceeds the memory budget by 40 megabytes.

PEOPLE / Sora: Developer presenting the benchmark | Malik: Developer reviewing the tradeoff

DO NOT HIDE A CONSTRAINT INSIDE A HEADLINE

Faster is incomplete when a resource budget matters. State the metric, conditions, comparison baseline, and constraint together. A colleague can acknowledge a genuine latency improvement while rejecting the claim that it settles the overall implementation decision.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which conditions are matched?

A Tail-latency percentiles

B Input and machine

C Production traffic and every user

D All future workloads

2. How far is B over the memory budget?

A 80 megabytes

B It is under budget

C 40 megabytes

D 15 megabytes

3. What is unmeasured?

A The budget difference

B Tail latency and production behavior

C The stated median values

D The stated memory values

Answers and explanations: [page 86](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

benchmark

Structured measurement under specified conditions.

IN USE run a benchmark

latency

Time taken for a request or operation.

IN USE measure latency

median

Middle value of an ordered set of measurements.

IN USE report median latency

tail latency

Slower-end behavior in a latency distribution.

IN USE measure tail latency

percentile

Value below which a stated percentage of observations falls.

IN USE report a latency percentile

p95

Ninety-fifth percentile of a measured distribution.

IN USE distinguish p95 from the median

throughput

Amount of work completed per unit of time.

IN USE measure throughput separately

memory footprint

Amount of memory used under the stated conditions.

IN USE compare the memory footprint

resource budget

Allowed limit for a specified resource.

IN USE respect the resource budget

headroom

Remaining capacity below a stated limit.

IN USE calculate memory headroom

overrun

Amount by which usage exceeds a limit.

IN USE quantify the budget overrun

baseline

Reference implementation or measurement for comparison.

IN USE use A as the baseline

Precision check

B reduces the median from 40 to 25 milliseconds: a 15-millisecond difference and a 37.5 percent reduction relative to A. It uses twice A's measured memory, but doubling memory is not the same as doubling speed.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

relative reduction

Decrease expressed as a fraction of the original value.

IN USE calculate the relative reduction

absolute difference

Direct numerical subtraction between comparable measurements.

IN USE report the absolute difference

tradeoff

Gain in one property accompanied by a cost in another.

IN USE explain the tradeoff

workload

Type and amount of work applied to a system.

IN USE describe the workload

sample

Set of observations used in an analysis.

IN USE document the sample

distribution

Pattern of observed values across a range.

IN USE inspect the latency distribution

outlier

Observation unusually distant from most other values.

IN USE investigate an outlier

measurement noise

Variation not attributable to the intended comparison alone.

IN USE account for measurement noise

controlled conditions

Specified conditions held consistent during comparison.

IN USE preserve controlled conditions

production behavior

Behavior under actual live operating conditions.

IN USE avoid assuming production behavior

bottleneck

Constraint limiting overall system performance.

IN USE identify the bottleneck

budget compliance

Whether measured usage satisfies the stated limit.

IN USE check budget compliance

Precision check

A uses 80 megabytes, leaving 40 below the 120-megabyte budget. B uses 160, exceeding it by 40. Neither median describes the unmeasured tail, and neither controlled result establishes production behavior.

Phrases you can actually use

C Notice the language / 1 of 2

NAME THE CONDITIONS

"Both runs use the same input and machine."

STATE B

"B has a twenty-five-millisecond median and uses one hundred sixty megabytes."

NAME THE BASELINE

"That is a thirty-seven-point-five percent reduction relative to A."

STATE THE CONSTRAINT

"The memory budget is one hundred twenty megabytes."

STATE A

"A has a forty-millisecond median and uses eighty megabytes."

REPORT THE ABSOLUTE GAIN

"B lowers the median by fifteen milliseconds."

REPORT THE COST

"B uses twice A's measured memory."

CALCULATE THE OVERRUN

"B exceeds that budget by forty megabytes."

06

Relative to A

Names the denominator for the percentage comparison.

By versus to

By gives the change; to gives the resulting value.

Twice

Compares B's memory with A's, not with the budget.

Phrases you can actually use

C Notice the language / 2 of 2

CALCULATE THE HEADROOM

"A remains forty megabytes below the limit."

LIMIT THE METRIC

"The median does not tell us the tail latency."

SEPARATE THROUGHPUT

"These figures do not establish throughput."

KEEP THE DECISION BOUNDED

"The measured gain does not remove the budget constraint."

AVOID A VAGUE VERDICT

"B is faster on this median measure, not automatically better overall."

NAME THE MISSING EVIDENCE

"We have not measured production behavior."

REQUEST A BALANCED SUMMARY

"Put the latency gain and memory cost together."

CLOSE WITHOUT OVERCLAIMING

"We have a documented tradeoff, not a production guarantee."

06

On this measure

Limits a faster claim to the reported metric.

Does not tell us

Prevents inference from median to unmeasured tail behavior.

Not automatically

Preserves the benefit while resisting an overall conclusion.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. What is B's median reduction relative to A?

- A** 100 percent
- B** 37.5 percent
- C** 60 percent
- D** 15 percent

2. Which budget statement is correct?

- A** B is 80 below budget
- B** A is 40 above budget
- C** A is 40 below and B is 40 above
- D** Both are within budget

3. Which claim exceeds the evidence?

- A** B has a lower measured median
- B** B uses more measured memory
- C** The benchmark used the same machine
- D** B improves every production request

4. Which summary is balanced?

- A** B is universally superior
- B** A is faster in the supplied test
- C** Memory no longer matters because latency improved
- D** B lowers median latency but exceeds the memory budget; tail and production remain unmeasured

Answers and explanations: [page 86](#).

Fifteen milliseconds faster, forty megabytes over budget

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

06

WORD + PHRASE BANK

budget compliance / benchmark / tradeoff / Production behavior / headroom / Tail latency / controlled conditions / relative reduction / memory footprint / absolute difference

- 01 **Sora:** The benchmark gives B a lower median: twenty-five milliseconds against A's forty. Both used the same input and machine, so the comparison conditions are matched.
- 02 **Malik:** That is useful ¹ _____ evidence. Put the memory figures beside it before we call B the better option. The resource budget is part of this decision.
- 03 **Sora:** A uses eighty megabytes and B uses one hundred sixty. The budget is one hundred twenty, so B does not fit the stated limit.
- 04 **Malik:** Right. Its ² _____ is twice A's and forty megabytes over budget. The latency improvement does not cancel that constraint.
- 05 **Sora:** I want the percentage wording right. The median drops by fifteen milliseconds from a forty-millisecond baseline, which is a thirty-seven-point-five percent reduction.
- 06 **Malik:** Yes, that is the ³ _____ relative to A. Keep the baseline explicit so nobody divides by twenty-five and reports a different quantity under the same label.
- 07 **Sora:** Could I say B is thirty-seven-point-five percent faster? That sounds more compact, but I can see it might blur latency with another speed measure.
- 08 **Malik:** Prefer the precise ⁴ _____ and percentage latency reduction. Those statements tell readers exactly which measurement changed without suggesting an unmeasured throughput result.
- 09 **Sora:** We also have no tail-latency data. The median does not tell us whether the slowest requests improved or became worse.
- 10 **Malik:** Correct. ⁵ _____ remains unmeasured. Do not turn a better middle value into a claim about every request or a specific percentile we did not record.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 55.

- 11 **Sora:** For A, the memory calculation goes the other way: eighty against a limit of one hundred twenty leaves forty megabytes available.
- 12 **Malik:** That is forty megabytes of ⁶ _____ under the stated budget. It is a different claim from saying A has more memory available under all possible workloads.
- 13 **Sora:** Should the summary mention production explicitly? These measurements come from the controlled comparison, and we have not collected live-environment results.
- 14 **Malik:** Yes. ⁷ _____ is not established by these figures. Name that limit instead of relying on readers to infer it from the word benchmark.
- 15 **Sora:** Then the headline should combine the gain and cost: lower measured median, higher memory, and a budget violation for B.
- 16 **Malik:** That presents the ⁸ _____ fairly. It acknowledges a real improvement without treating one favorable number as a complete engineering decision.
- 17 **Sora:** I will not ask for the budget to disappear from the table. If someone proposes a different limit, that would be another decision, not a property of this measurement.
- 18 **Malik:** Exactly. Current ⁹ _____ is clear: A is below the limit and B is above it. A hypothetical future budget should not rewrite today's comparison.
- 19 **Sora:** My final summary will report both values, the fifteen-millisecond decrease, the memory difference, and the missing tail and production measurements.
- 20 **Malik:** Good. Keep the ¹⁰ _____ visible alongside that summary. We can make a reasoned next decision without pretending the benchmark answers questions it never measured.

06

ANSWERS / EXPLANATIONS: PAGE 87

1. benchmark | 2. memory footprint | 3. relative reduction | 4. absolute difference | 5. Tail latency | 6. headroom | 7. Production behavior | 8. tradeoff | 9. budget compliance | 10. controlled conditions

Report both sides of the comparison

F Say it

Use the supplied numbers to complete the benchmark summary. Units are printed in each sentence.

Complete the four gaps. Use each bank entry once.

06

WORD + PHRASE BANK

160 / 25 / 15 / 40

Developer: "B's median is ¹ _____ milliseconds."

Reviewer: "The median decreases by ² _____ milliseconds relative to A."

Developer: "B uses ³ _____ megabytes of memory."

Reviewer: "B exceeds the budget by ⁴ _____ megabytes."

ANSWERS / EXPLANATIONS: PAGE 87

1. 25 | 2. 15 | 3. 160 | 4. 40

Rehearse with a fixed script

1. Check your answers, then read turns 1-10 with a partner using the corrected words.
2. Switch roles for turns 11-20. Stress words that distinguish facts from conditions or estimates.
3. Complete and check the four-line exchange. Read it twice, switching roles and keeping the printed facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

The timeout did not answer the business question

A Read the situation

Explain an uncertain request outcome and request a retry contract before risking a second reservation.

A fictional reservation client times out before receiving a reply. The service may already have created the reservation. A proposed automatic retry would submit a new reservation request, and the service's duplicate-handling behavior is unspecified. Developers Imani and Leo discuss the risk. No production change is authorized. They need to clarify outcome reconciliation and retry semantics, then test agreed behavior in a suitable controlled environment.

PEOPLE / Imani: Client developer | Leo: Service developer

07

DISTINGUISH NOT KNOWING FROM KNOWING FAILURE

Timeout language can silently collapse uncertainty into failure. Say what the client observed and what the service may have done. This makes a request for an explicit retry contract sound like necessary coordination rather than unnecessary caution.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What did the client observe?

- A A verified cancellation
- C Confirmed rejection before processing

- B No reply before its timeout
- D Confirmed creation twice

2. What might the service have done?

- A Created the reservation already
- C Guaranteed cancellation

- B Definitely done nothing
- D Automatically merged duplicates

3. What is unspecified?

- A Whether the client timed out
- C Whether production authorization exists

- B Whether the retry is proposed as new
- D Duplicate-handling behavior for the proposed retry

Answers and explanations: [page 88](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

timeout

End of a waiting period without the required response.

IN USE observe a timeout

retry

Repeated attempt after an earlier uncertain or unsuccessful attempt.

IN USE define retry behavior

request identity

Identifier or meaning used to distinguish one operation from another.

IN USE preserve request identity

idempotency

Property allowing repeated equivalent requests without additional intended effects.

IN USE specify idempotency semantics

idempotency key

Identifier used under an agreed contract to recognize repeated intent.

IN USE define an idempotency key

duplicate request

Repeated request that may represent the same intended operation.

IN USE detect a duplicate request

duplicate side effect

Additional effect caused by repeating an operation.

IN USE prevent a duplicate side effect

shared state

Data or condition used by multiple participants or operations.

IN USE reconcile shared state

uncertain outcome

Situation where the operation's final result is not known.

IN USE report an uncertain outcome

acknowledgment

Response indicating receipt or completion as defined by the protocol.

IN USE distinguish an acknowledgment

reconciliation

Checking records to establish and align the actual outcome.

IN USE request outcome reconciliation

at-least-once delivery

Delivery model allowing one or more deliveries of a message.

IN USE discuss at-least-once delivery

Precision check

A timeout describes the client's wait, not proof that the service did nothing. Submitting a new reservation could create another business effect if the first request already succeeded and duplicate handling is not defined.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

at-most-once delivery

Delivery model allowing no more than one delivery, with possible loss.

IN USE distinguish at-most-once delivery

deduplication

Recognition and handling of repeated requests or records.

IN USE define deduplication behavior

race condition

Behavior depending on timing among concurrent operations.

IN USE investigate a race condition

concurrency

Overlapping progress of multiple operations.

IN USE account for concurrency

atomic operation

Operation treated as an indivisible state change under its model.

IN USE define the atomic operation

transaction boundary

Scope within which specified changes are coordinated.

IN USE clarify the transaction boundary

backoff

Delay strategy between repeated attempts.

IN USE configure retry backoff

retry budget

Limit on permitted repeated attempts under a policy.

IN USE set a retry budget

correlation identifier

Identifier linking related events across components.

IN USE retain the correlation identifier

status lookup

Query intended to retrieve an operation's recorded state.

IN USE define a status lookup

sandbox

Controlled environment separated from live operations.

IN USE test in a sandbox

authorization

Permission for a specified action or change.

IN USE obtain production authorization

Precision check

An idempotency key is not a magic guarantee. Its scope, reuse rules, retention, responses, and service behavior need a contract. Backoff changes timing; it does not by itself resolve an uncertain outcome or prevent duplication.

Phrases you can actually use

C Notice the language / 1 of 2

STATE THE OBSERVATION

"The client timed out before receiving a reply."

AVOID A FALSE CONCLUSION

"No reply does not prove no reservation."

NAME THE RISK

"That could create a duplicate business effect."

AVOID ASSUMING SUPPORT

"Duplicate handling is not specified."

PRESERVE UNCERTAINTY

"The service may already have created the reservation."

IDENTIFY THE PROPOSAL

"The retry would submit a new reservation request."

ASK FOR THE CONTRACT

"How does the service recognize repeated intent?"

CLARIFY IDENTITY

"Would this carry the same request identity or a new one?"

May already have

Expresses a possible completed action whose result is not yet known.

Does not prove

Blocks an invalid inference from silence to failure.

Could create

Names a risk without falsely asserting duplication already occurred.

Phrases you can actually use

C Notice the language / 2 of 2

REQUEST RECONCILIATION

"Can we establish the outcome of the first request?"

QUALIFY THE KEY

"An idempotency key needs agreed service behavior."

REQUEST A CONTROLLED CHECK

"Test the agreed behavior in a suitable sandbox."

KEEP THE DESIGN PROVISIONAL

"Automatic retry remains a proposal."

SEPARATE TIMING AND SEMANTICS

"Backoff does not itself prevent duplicates."

STATE THE PERMISSION BOUNDARY

"No production change is authorized."

NAME THE MISSING EVIDENCE

"We do not know whether the first reservation exists."

CLOSE WITH THE DEPENDENCY

"Define the contract before claiming retry safety."

Same ... or new

Exposes request identity as a design decision.

By itself

Limits what backoff or a key can guarantee alone.

Before claiming

Makes a confidence statement dependent on missing contract evidence.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which statement follows from the timeout?

- A** Two reservations definitely exist
- B** The service canceled the request
- C** The client did not receive a timely reply
- D** No reservation exists

2. Why is a new reservation request risky?

- A** Backoff guarantees a second failure
- B** The first may already have created a reservation
- C** Every retry always duplicates
- D** The client has confirmed cancellation

3. What does an idempotency key require?

- A** No server participation
- B** Automatic universal guarantees
- C** Agreed service semantics
- D** Only a convenient label

4. Which next step is appropriate?

- A** Enable the retry in production immediately
- B** Assume timeout means rejection
- C** Promise exactly one reservation without evidence
- D** Clarify reconciliation and retry behavior, then test it outside production

[Answers and explanations: page 88.](#)

The timeout did not answer the business question

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

shared state / timeout / deduplication / Backoff / request identity / reconciliation / uncertain outcome / status lookup / idempotency key / authorization

07

- 01 **Imani:** The reservation client timed out before it received a reply. I have a proposal to retry automatically, but it currently submits the second attempt as a new reservation.
- 02 **Leo:** We need to preserve the ¹ _____ of the first attempt. The service may already have created it, so the timeout is not proof that nothing happened.
- 03 **Imani:** That is the gap in my current explanation. I described the request as failed, when what I actually know is that the client stopped waiting.
- 04 **Leo:** Exactly. A ² _____ is an observation at the client. We need separate evidence about the service state before turning that into a business conclusion.
- 05 **Imani:** If the first reservation exists, the new request might create another. We do not have documented behavior saying the service will recognize it as the same intent.
- 06 **Leo:** Then ³ _____ must remain unspecified in the proposal. We cannot promise the second request will be merged or ignored just because the input looks similar.
- 07 **Imani:** Would adding a request key solve it? I have heard teams describe keys as making retries safe, but I do not know this service's rules.
- 08 **Leo:** An ⁴ _____ needs agreed semantics on the service side. A new field alone does not tell us its scope, reuse rules, or what response a repeat receives.
- 09 **Imani:** We should also be clear whether the retry represents the original operation or a genuinely new booking. Giving it a new identity could contradict our intended behavior.
- 10 **Leo:** Yes. Define the ⁵ _____ and how the service interprets it. That distinction belongs in the contract, not in an assumption hidden inside the client's retry loop.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 64.

- 11 **Imani:** Before another attempt, could we establish the first result through a supported status check? I do not want to invent an endpoint that this service may not have.
- 12 **Leo:** Request an agreed ⁶ _____ or another supported reconciliation process. Its availability and meaning need confirmation; we have not established either in this discussion.
- 13 **Imani:** Someone suggested adding a delay between attempts. That could reduce repeated traffic, but it would not tell us whether the first reservation already exists.
- 14 **Leo:** Correct. ⁷ _____ is a timing policy, not a substitute for outcome and duplicate-handling semantics. A slower duplicate is still a possible duplicate.
- 15 **Imani:** I will revise the proposal around those unresolved questions. There is no permission to enable the automatic retry in production while we investigate.
- 16 **Leo:** Keep that ⁸ _____ boundary visible. Once the contract is defined, we can plan a controlled check without changing live reservation behavior.
- 17 **Imani:** The test should cover a reply arriving too late after the service may have completed the operation, not only an immediate rejection before any work.
- 18 **Leo:** That will address the relevant ⁹ _____ uncertainty. Use a suitable sandbox and define the expected outcome before interpreting the test as evidence.
- 19 **Imani:** My summary is no timely reply, first outcome unknown, proposed retry currently new, duplicate handling unspecified, and no production change approved.
- 20 **Leo:** Good. We need ¹⁰ _____ and an explicit retry contract before claiming safety. That is the concrete next discussion, not a guarantee that an extra attempt will be harmless.

07

ANSWERS / EXPLANATIONS: PAGE 89

1. uncertain outcome | 2. timeout | 3. deduplication | 4. idempotency key | 5. request identity | 6. status lookup | 7. Backoff | 8. authorization | 9. shared state | 10. reconciliation

Do not turn silence into failure

F Say it

Complete the retry discussion using the stated uncertainty and permission boundary.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

authorized / timeout / unspecified / exist

07

Developer: "The client observed a ¹ _____."

Reviewer: "The first reservation may already ² _____."

Developer: "Duplicate handling remains ³ _____."

Reviewer: "No production change is ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 89

1. timeout | 2. exist | 3. unspecified | 4. authorized

Rehearse with a fixed script

1. Check your answers, then read turns 1-10 with a partner using the corrected words.
2. Switch roles for turns 11-20. Stress words that distinguish facts from conditions or estimates.
3. Complete and check the four-line exchange. Read it twice, switching roles and keeping the printed facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

R18 is deployed, but the release decision is not made

A Read the situation

Hand over environment, flag state, migration limits, and decision ownership without conflating deployment with production readiness.

Release R18 is deployed in staging. Its new search feature is disabled by a feature flag. A migration adds an optional field, but compatibility with the previous application version has not been tested. Developer Chen hands the status to release reviewer Lee. The production decision remains with Lee and has not been made. Neither the optional field nor the staging deployment establishes that an older application will safely work with the migrated data.

PEOPLE / Chen: Developer handing over R18 | Lee: Release reviewer

08

SAY DONE WITH AN OBJECT ATTACHED

Deployed, enabled, tested, and approved describe different milestones. Attach each status to an environment, feature, or specific check. This reduces pressure on the next person to infer readiness from a single broad word such as ready or complete.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Where is R18 deployed?

- A Production
- C Nowhere

- B Every customer environment
- D Staging

2. What is the new search feature's state?

- A Enabled for all users
- C Proven compatible with every client

- B Deleted from the release
- D Disabled by a feature flag

3. Which check is unresolved?

- A Compatibility with the previous application version
- C Whether the field is optional

- B Whether the staging deployment occurred
- D Who owns the production review

Answers and explanations: page 90.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

release candidate

Version proposed for release pending the relevant checks.

IN USE review the release candidate

deployment

Placement of software into a specified environment.

IN USE report the deployment

staging

Preproduction environment used for relevant checks.

IN USE deploy to staging

production

Live environment serving actual operational users or workloads.

IN USE authorize production release

feature flag

Control that enables or disables specified behavior.

IN USE verify the feature flag

disabled state

Condition in which a feature is not enabled.

IN USE preserve the disabled state

rollout

Process of making a change available to intended users.

IN USE plan the rollout

migration

Change to stored data or its structure.

IN USE review the migration

optional field

Field that may be absent under the applicable rules.

IN USE add an optional field

schema compatibility

Ability of components to work with the relevant data structure.

IN USE verify schema compatibility

previous version

Earlier application version relevant to the comparison.

IN USE test the previous version

rollback

Return to an earlier application or system state under a defined plan.

IN USE assess rollback readiness

Precision check

R18 is deployed in staging, and search is disabled. Those are two separate facts. They do not imply production deployment, enabled search, completed compatibility testing, or Lee's approval to proceed.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

forward compatibility

Ability to work with relevant newer data or interfaces.

IN USE check forward compatibility

backward compatibility

Ability to preserve supported use by earlier consumers.

IN USE verify backward compatibility

release gate

Required condition before advancing a release.

IN USE satisfy the release gate

sign-off

Explicit approval at a stated decision point.

IN USE obtain release sign-off

go/no-go decision

Decision whether to proceed under the relevant release process.

IN USE record the go/no-go decision

handoff

Transfer of relevant status, evidence, and responsibility.

IN USE prepare the release handoff

known limitation

Identified restriction or unresolved boundary.

IN USE list known limitations

verification matrix

Organized record of combinations and checks to be verified.

IN USE update the verification matrix

environment boundary

Limit separating claims about different execution settings.

IN USE preserve the environment boundary

data compatibility

Ability to interpret and use the relevant stored data correctly.

IN USE test data compatibility

release owner

Person responsible for the specified release decision.

IN USE name the release owner

readiness claim

Statement that required conditions for proceeding are met.

IN USE qualify the readiness claim

Precision check

An optional new field may look low risk, but its optional status does not prove an older application can handle the changed data or schema. Rollback readiness requires evidence about the earlier version and the actual post-migration state.

Phrases you can actually use

C Notice the language / 1 of 2

IDENTIFY THE RELEASE

"This handoff concerns R18."

NAME THE FEATURE STATE

"New search remains disabled by its flag."

DESCRIBE THE DATA CHANGE

"The migration adds an optional field."

AVOID A SHORTCUT

"Optional does not prove compatibility."

NAME THE ENVIRONMENT

"R18 is deployed in staging."

SEPARATE DEPLOYMENT AND USE

"Deployed does not mean enabled."

STATE THE MISSING CHECK

"Compatibility with the previous application version is untested."

NAME THE DECISION OWNER

"Lee owns the production review."

08

Remains disabled

Preserves feature state despite completed deployment.

Against

Connects the previous application to the data state it must handle.

Not production sign-off

Prevents a handoff from being mistaken for permission.

Phrases you can actually use

C Notice the language / 2 of 2

KEEP THE DECISION OPEN

"The production decision has not been made."

REQUEST THE EXACT EVIDENCE

"Check the previous application against the migrated state."

IDENTIFY THE PENDING ITEM

"The compatibility check remains open."

PRESERVE THE BOUNDARY

"Do not describe search as live."

LIMIT THE READINESS CLAIM

"This is a staging status, not production sign-off."

SEPARATE ROLLBACK ASSUMPTIONS

"An older build alone does not establish rollback readiness."

ASK FOR A TRACEABLE DECISION

"Record the review outcome explicitly."

CLOSE WITH STATUS

"Staging deployed, search disabled, compatibility pending, decision with Lee."

08

Has not been made

Makes the unresolved decision explicit.

Alone

Limits what an available earlier build establishes.

With Lee

Assigns decision ownership without predicting approval.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which status is accurate?

- A** R18 has full production sign-off
- B** The old application is verified compatible
- C** R18 is in staging with search disabled
- D** Search is live in production

2. Why does optional not settle compatibility?

- A** Optional fields are always incompatible
- B** The migration never ran anywhere
- C** Lee has already rejected release
- D** The previous application's behavior with migrated data is untested

3. Which rollback claim is justified?

- A** An older build guarantees safe rollback
- B** Staging deployment proves data reversibility
- C** A disabled flag automatically undoes the migration
- D** Readiness remains unverified without the relevant compatibility evidence

4. Who owns the unresolved production decision?

- A** The feature flag itself
- B** The migration tool
- C** Lee
- D** Every staging user

Answers and explanations: [page 90](#).

R18 is deployed, but the release decision is not made

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

rollback / readiness claim / environment boundary / feature flag / Schema compatibility / data compatibility / release gate / handoff / known limitation / go/no-go decision

08

- 01 **Chen:** I am handing over R18 for your release review. It is deployed in staging, and the new search feature remains disabled by its flag.
- 02 **Lee:** Keep that ¹ _____ explicit in the summary. I do not want deployed shortened into live when someone forwards this message.
- 03 **Chen:** Agreed. The deployment is complete in staging, but the feature is not enabled there. I will report those as separate statuses.
- 04 **Lee:** Good. The ² _____ state matters independently of where the code is installed. Neither fact is a production decision.
- 05 **Chen:** The migration adds an optional field. I initially wrote backward compatible in the handoff, but we have not tested the previous application version against the changed state.
- 06 **Lee:** Then replace that with ³ _____. Optional describes the field, not the older application's demonstrated behavior. We need to preserve the missing check.
- 07 **Chen:** I will say compatibility with the previous application version remains untested. That is more accurate than predicting it from the small size of the change.
- 08 **Lee:** Yes. ⁴ _____ is a claim about interaction with the changed structure. A field being optional does not automatically establish every consumer's handling.
- 09 **Chen:** Someone may ask whether we can simply return to the previous build if production review uncovers a problem. We have not verified that path either.
- 10 **Lee:** Do not claim ⁵ _____ readiness from the existence of an older build alone. Its behavior against the post-migration state is part of the unresolved question.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 73.

- 11 **Chen:** Should I list that as a request to check the prior application with the migrated data, rather than as a request merely to locate the old build?
- 12 **Lee:** Exactly. The relevant ⁶ _____ evidence needs the real combination under review. Availability of a file is not the same as verified operation.
- 13 **Chen:** I will keep the review outcome separate from the handoff. Sending you this status does not mean I have permission to enable search or proceed to production.
- 14 **Lee:** Correct. I own the ⁷ _____ for production, and it has not been made. Please do not phrase my pending review as expected approval.
- 15 **Chen:** The summary can list R18, staging, search disabled, optional-field migration, untested prior-version compatibility, and your decision ownership. That should be enough to avoid an ambiguous ready label.
- 16 **Lee:** That is a useful ⁸ _____. It gives the next person concrete facts and an open item instead of a broad status that means different things to different readers.
- 17 **Chen:** Once the missing check is performed, its actual result should be added. Until then, I will not mark compatibility complete because the staging deployment succeeded.
- 18 **Lee:** Right. Keep the ⁹ _____ distinct from a completed deployment step. The relevant review still needs its evidence and an explicit decision.
- 19 **Chen:** My final line will be staging deployed, search disabled, compatibility pending, production decision with Lee. There will be no claim that search is live.
- 20 **Lee:** That supports an accurate ¹⁰ _____. We can review what is actually known without confusing a status update with authorization to release.

08

ANSWERS / EXPLANATIONS: PAGE 91

1. environment boundary | 2. feature flag | 3. known limitation | 4. Schema compatibility | 5. rollback | 6. data compatibility | 7. go/no-go decision | 8. handoff | 9. release gate | 10. readiness claim

Four facts for the release handoff

F Say it

Complete the R18 handoff. Do not upgrade any pending check or decision to completed status.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

Lee / untested / staging / disabled

08

Developer: "R18 is deployed in ¹ _____."

Reviewer: "New search remains ² _____."

Developer: "Prior-version compatibility is ³ _____."

Reviewer: "The production decision remains with ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 91

1. staging | 2. disabled | 3. untested | 4. Lee

Rehearse with a fixed script

1. Check your answers, then read turns 1-10 with a partner using the corrected words.
2. Switch roles for turns 11-20. Stress words that distinguish facts from conditions or estimates.
3. Complete and check the four-line exchange. Read it twice, switching roles and keeping the printed facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

01 Briefing and language checks

Clarifying interface contracts

A Briefing checks / page 4

1. What does omitting nickname do under the current contract?

B. Leaves the stored value unchanged

The current contract explicitly preserves the value when nickname is omitted.

2. What does an empty string do?

D. Clears the nickname

The empty string has a defined clearing meaning distinct from omission.

3. What remains undefined?

A. The meaning of an explicit null

The proposed client sends null, whose meaning the contract has not specified.

D Language checks / page 9

1. Which comment identifies the real gap?

C. The null case needs an agreed meaning

Only null lacks a specified meaning in this particular contract.

2. Which pair must remain distinct?

B. Omission and an empty string

Omission preserves the value, whereas an empty string clears it.

3. What would justify calling null a clearing command?

D. An explicit agreed contract change

A new interpretation needs agreement rather than an unsupported inference.

4. Which test plan matches the discussion?

A. Check each defined input state and resolve null before asserting its effect

Distinct input states require distinct expectations, with null still unresolved.

01 Conversation answers

Clarifying interface contracts

E Dialogue gaps

Pages 10-11

1. omitted field. Turn 2. An omitted field is absent and preserves the stored nickname under the stated contract.

2. optional field. Turn 4. Optional field describes allowed absence, not the meaning or acceptability of an explicit null.

3. empty string. Turn 6. An empty string has the specified effect of clearing the nickname, unlike omission.

4. interface contract. Turn 8. The interface contract is the agreement that must specify the missing null behavior before implementation.

5. request payload. Turn 10. Request payload identifies the actual submitted data, which may differ from an assumed client representation.

6. field presence. Turn 12. Field presence distinguishes an absent nickname property from a present property carrying a value.

7. validation rule. Turn 14. The validation rule must reflect an agreed input policy rather than preempt the unresolved decision.

8. contract test. Turn 16. A contract test checks the agreed interaction across the client-service boundary rather than only the visible form.

9. backward compatibility. Turn 18. Backward compatibility concerns preserving supported behavior for existing clients, including the established omission rule.

10. acceptance criterion. Turn 20. An acceptance criterion must name the agreed input and effect before implementation can be judged complete.

F Transfer conversation

Page 12

1. unchanged

Developer: "An omitted nickname leaves the value unchanged."

Omission explicitly preserves the existing nickname under the current contract.

2. clears

Reviewer: "An empty string clears the nickname."

The empty string is the defined clearing input in this interface.

3. unspecified

Developer: "The meaning of null remains unspecified."

No agreed effect for null is supplied in the scenario.

4. nullable

Reviewer: "Optional does not automatically mean nullable."

Optional concerns absence, while nullable concerns allowed explicit values.

02 Briefing and language checks

Debugging from reproducible evidence

A Briefing checks / page 13

1. Which input fails?
B. A three-row file containing an empty date
Only the stated three-row example is established as failing.
2. What does the successful comparison change?
B. Both row count and content
Removing the row changes the number of rows and their content.
3. What does the stack trace establish?
D. The observed failure path reaches date parsing
The trace locates part of the failure path but does not isolate the cause.

D Language checks / page 18

1. Which conclusion is supported?
B. The observed failing path reaches date parsing
The trace supports a location statement, not the broader causal claims.
2. Which proposed comparison is strongest?
B. Keep three rows and other content fixed while varying the suspect date
Holding other factors fixed better isolates the condition being tested.
3. Which phrase avoids overstating evidence?
D. The empty date is a working hypothesis
A working hypothesis accurately labels the explanation as still requiring evidence.
4. What should the handoff preserve?
A. The failing fixture, conditions, and trace
Reproducible evidence needs the original input and execution context.

02 Conversation answers

02

Debugging from reproducible evidence

E Dialogue gaps

Pages 19-20

1. reproducible example. Turn 2. A reproducible example provides the input and conditions needed to observe the reported failure again.

2. hypothesis. Turn 4. Hypothesis labels the empty-date explanation as plausible but not yet isolated by a controlled test.

3. confounding variable. Turn 6. A confounding variable is another changed factor that prevents this comparison from isolating the date condition.

4. controlled comparison. Turn 8. Controlled comparison holds the other relevant conditions fixed while varying the suspected date condition.

5. input fixture. Turn 10. Input fixture names the prepared test data, which must remain available to reproduce the original failure.

6. stack trace. Turn 12. The stack trace records the observed call path around the error but does not alone establish its root cause.

7. failure signature. Turn 14. Failure signature distinguishes the original error pattern from a different failure in the comparison run.

8. deterministic behavior. Turn 16. Deterministic behavior concerns consistent results under the same relevant conditions, which repeated checks can investigate.

9. nonproduction environment. Turn 18. A nonproduction environment supports the investigation without treating this dialogue as permission to test on live workloads.

10. causal claim. Turn 20. A causal claim needs stronger evidence than the current confounded comparison, so the proposed cause remains qualified.

F Transfer conversation

Page 21

1. three

Developer: "The failing file contains three rows."

The supplied failing fixture has three rows, including one empty date.

2. two

Reviewer: "The successful file contains two rows."

Removing the suspect row leaves the stated successful two-row file.

3. hypothesis

Developer: "The empty date remains a hypothesis."

The current comparison has not isolated the date as the cause.

4. constant

Reviewer: "Keep the other conditions constant in the next comparison."

Holding other conditions constant helps isolate the suspected date effect.

03 Briefing and language checks

Giving actionable code-review feedback

03

A Briefing checks / page 22

1. What is the pull request titled?

B. Rename export helper

The stated title describes a helper rename, not a notification-policy change.

2. What additional change appears in the diff?

B. Notification moves outside the success branch

The changed call location permits notification after an unsuccessful export.

3. What part does Priya accept?

D. The naming change

Priya accepts the rename while objecting to the changed notification behavior.

D Language checks / page 27

1. Which comment is actionable?

A. Restore the success-only notification and test that failure sends no completion notice

The correct comment names the behavior, requested change, and verification.

2. Why is this not only a refactor?

A. Observable notification behavior changes

A behavior-preserving refactor would not add completion notices after failure.

3. What should the failure test assert?

B. No completion notice is sent

The defect concerns a false success message after export failure.

4. Which statement preserves Priya's position?

A. The rename is acceptable; the notification behavior needs correction

Priya distinguishes an acceptable naming change from an unresolved behavioral defect.

03 Conversation answers

Giving actionable code-review feedback

03

E Dialogue gaps

Pages 28-29

1. rename. Turn 2. Rename identifies the accepted identifier change, which does not explain or justify the additional behavioral change.

2. control flow. Turn 4. Control flow determines whether a failed export can reach the notification call after it moves outside the branch.

3. side effect. Turn 6. Side effect refers to the externally visible notification rather than the helper's new name.

4. failure path. Turn 8. The failure path must be exercised to check that unsuccessful exports do not trigger a completion notice.

5. review comment. Turn 10. A review comment is actionable when it identifies the changed location, effect, and requested correction.

6. review scope. Turn 12. Review scope should reflect the actual remaining changes rather than rely solely on the original rename title.

7. false success. Turn 14. False success is the completion indication produced even though the export operation failed.

8. patch revision. Turn 16. A patch revision is an updated proposed change, which still needs review rather than automatic approval.

9. regression test. Turn 18. A regression test should detect recurrence of the known false-notification defect rather than merely execute the helper.

10. verification evidence. Turn 20. Verification evidence supports the revised behavior and is needed before the reviewer can resolve the blocking concern.

F Transfer conversation

Page 30

1. name

Reviewer: "The helper name is acceptable."

Priya explicitly accepts the naming change in the pull request.

2. success

Author: "The notification must remain on the success path."

Completion notification is intended only for a successful export here.

3. failure

Reviewer: "Add a test for the failure path."

The regression occurs when export fails yet a completion notice appears.

4. approval

Author: "The current feedback is not yet approval."

The blocking behavior issue still requires correction and verification.

04 Briefing and language checks

Requesting dependency evidence

04

A Briefing checks / page 31

1. Which dependency does the application use directly?

B. ParcelKit

The application uses ParcelKit directly, while TextCore is introduced through ParcelKit.

2. What changes indirectly?

C. The resolved TextCore version

The proposed ParcelKit upgrade changes the TextCore version selected in the dependency resolution.

3. What does the note establish?

A. A parsing entry point is deprecated

Deprecation is stated, but usage, removal, and replacement compatibility remain unverified.

D Language checks / page 36

1. Which statement is accurate?

A. TextCore is transitive through ParcelKit

TextCore is brought into the application through its direct dependency ParcelKit.

2. What should be checked first about the selected version?

B. The resolved dependency record

The resolved record establishes the concrete TextCore version selected by this upgrade.

3. What does deprecated not establish?

B. That the entry point has been removed

Deprecation and removal are different statuses, and removal has not been established.

4. Which compatibility claim is justified now?

D. Compatibility remains unverified

Usage and replacement behavior have not yet been checked.

04 Conversation answers

Requesting dependency evidence

E Dialogue gaps

Pages 37-38

- 1. transitive dependency.** Turn 2. TextCore is a transitive dependency because the application receives it through ParcelKit rather than using it directly here.
- 2. lockfile.** Turn 4. The lockfile records resolved dependency selections and is more specific than a top-level version constraint alone.
- 3. deprecation.** Turn 6. Deprecation marks an interface for discouragement or transition but does not alone establish removal or current usage.
- 4. call site.** Turn 8. A call site shows where the affected parsing interface is invoked, which must be checked rather than assumed.
- 5. migration guide.** Turn 10. The migration guide provides transition details that a similar replacement name cannot establish.
- 6. semantic versioning.** Turn 12. Semantic versioning provides specific conventions only when the relevant project actually adopts and follows that policy.
- 7. dependency tree.** Turn 14. The dependency tree makes the ParcelKit-to-TextCore relationship visible so indirect impact is included in the review.
- 8. public interface.** Turn 16. The public interface is the exposed parsing surface whose supported behavior and affected callers need verification.
- 9. upgrade evidence.** Turn 18. Upgrade evidence combines actual versions, affected usage, documentation, and relevant checks rather than a broad assertion of safety.
- 10. compatibility.** Turn 20. Compatibility remains unverified until the relevant consumers and behavior have been checked against the proposed versions.

F Transfer conversation

Page 39

1. direct

Developer: "ParcelKit is the direct dependency."

The application uses ParcelKit directly under the stated dependency relationship.

2. transitive

Reviewer: "TextCore is the transitive dependency."

TextCore is introduced through ParcelKit rather than directly here.

3. deprecated

Developer: "The parsing entry point is marked deprecated."

The change note establishes deprecation, not removal or usage.

4. unverified

Reviewer: "Replacement compatibility remains unverified."

The relevant behavior and affected callers have not yet been checked.

04

05 Briefing and language checks

Explaining what a test actually exercises

A Briefing checks / page 40

1. What does the storage double currently do?

A. Always succeeds

The supplied test double is configured only for successful storage.

2. What did the previous bug show?

C. Success after storage rejected the save

The prior defect was a false success indication after rejection.

3. What remains separately untested?

C. The real storage adapter

The scenario explicitly states that the real adapter has not been tested.

D Language checks / page 45

1. Why does the current test miss the prior bug?

C. Its double never rejects

The failing condition required to expose the bug is absent from the fixture.

2. Which addition targets the defect?

C. Configure rejection and assert no success message

The new case must trigger rejection and verify the missing user-visible boundary.

3. What does the rejection unit test not establish?

A. Correct behavior of the real storage adapter

The test still substitutes for the real adapter rather than exercising it.

4. Which report is precise?

B. Success-path coverage exists; rejection and real-adapter coverage need work

The report separates existing coverage from two distinct unresolved areas.

05 Conversation answers

Explaining what a test actually exercises

E Dialogue gaps

Pages 46-47

- 1. happy path.** Turn 2. The happy path is the successful execution route supplied by the always-successful storage double.
- 2. test double.** Turn 4. The test double replaces real storage and must be configured to produce the rejection needed by this test.
- 3. negative assertion.** Turn 6. A negative assertion checks that the unwanted success message does not appear after the simulated rejection.
- 4. Interaction verification.** Turn 8. Interaction verification checks dependency calls but does not alone establish the correct user-visible result.
- 5. integration test.** Turn 10. An integration test would exercise the relevant connected components, whereas this case still replaces the real storage adapter.
- 6. coverage gap.** Turn 12. The coverage gap for the real adapter remains open even if the simulated rejection unit case passes.
- 7. arrange.** Turn 14. Arrange is the preparation stage that configures the rejection condition before the save behavior is executed.
- 8. success response.** Turn 16. Success response is the outcome configured in the existing test, so its evidence is limited to that path.
- 9. behavioral verification.** Turn 18. Behavioral verification checks the observed application result, while retaining the boundary imposed by the substituted dependency.
- 10. unit test.** Turn 20. A unit test can cover the application's rejection handling without proving the behavior of the real storage integration.

F Transfer conversation

Page 48

1. succeeds

Developer: "The existing double always succeeds."

The current fixture supplies only successful storage outcomes to the save operation.

2. rejection

Reviewer: "The new case must trigger rejection."

Rejection is the missing condition that exposed the previous false-success bug.

3. success

Developer: "The assertion should prevent a false success message."

The known defect incorrectly displayed success after storage rejected the save.

4. untested

Reviewer: "The real adapter remains untested."

Simulated rejection coverage does not exercise the real storage adapter.

05

06 Briefing and language checks

Comparing performance tradeoffs fairly

A Briefing checks / page 49

1. Which conditions are matched?

B. Input and machine

The benchmark explicitly uses the same input and machine for both implementations.

2. How far is B over the memory budget?

C. 40 megabytes

B uses 160 megabytes against a budget of 120, exceeding it by forty.

3. What is unmeasured?

B. Tail latency and production behavior

Only the benchmark median and memory figures are supplied, not tail or production results.

D Language checks / page 54

1. What is B's median reduction relative to A?

B. 37.5 percent

Fifteen divided by the forty-millisecond baseline equals 0.375, or 37.5 percent.

2. Which budget statement is correct?

C. A is 40 below and B is 40 above

The 120-megabyte limit lies forty above A and forty below B.

3. Which claim exceeds the evidence?

D. B improves every production request

Production behavior and every-request outcomes have not been measured.

4. Which summary is balanced?

D. B lowers median latency but exceeds the memory budget; tail and production remain unmeasured

The summary combines the measured benefit, resource constraint, and remaining evidence limits.

06 Conversation answers

Comparing performance tradeoffs fairly

E Dialogue gaps

Pages 55-56

- 1. benchmark.** Turn 2. Benchmark identifies the controlled measurement, whose conditions limit the scope of the comparison.
- 2. memory footprint.** Turn 4. Memory footprint is the measured memory usage, which is 160 megabytes for B against a 120-megabyte budget.
- 3. relative reduction.** Turn 6. Relative reduction divides the fifteen-millisecond decrease by A's forty-millisecond baseline, yielding 37.5 percent.
- 4. absolute difference.** Turn 8. Absolute difference is the direct fifteen-millisecond reduction, which can be reported alongside the relative latency reduction.
- 5. Tail latency.** Turn 10. Tail latency concerns slower-end observations and cannot be inferred from the supplied median alone.
- 6. headroom.** Turn 12. Headroom is the forty-megabyte difference between A's eighty-megabyte usage and the 120-megabyte limit.
- 7. Production behavior.** Turn 14. Production behavior remains unmeasured, so the controlled benchmark cannot establish live-system outcomes.
- 8. tradeoff.** Turn 16. Tradeoff connects B's lower median latency with its higher memory use and failure to meet the stated budget.
- 9. budget compliance.** Turn 18. Budget compliance is evaluated against the stated 120-megabyte limit, which B exceeds regardless of its latency gain.
- 10. controlled conditions.** Turn 20. Controlled conditions specify the matched input and machine and prevent the result being generalized without further evidence.

F Transfer conversation

Page 57

1. 25

Developer: "B's median is 25 milliseconds."

The measured median for implementation B is twenty-five milliseconds.

2. 15

Reviewer: "The median decreases by 15 milliseconds relative to A."

Forty minus twenty-five gives a fifteen-millisecond absolute decrease.

3. 160

Developer: "B uses 160 megabytes of memory."

The supplied memory measurement for implementation B is 160 megabytes.

4. 40

Reviewer: "B exceeds the budget by 40 megabytes."

One hundred sixty minus the 120-megabyte budget leaves forty over.

07 Briefing and language checks

Discussing retries and shared state

A Briefing checks / page 58

1. What did the client observe?

B. No reply before its timeout

A timeout establishes lack of a timely reply, not the service's final action.

2. What might the service have done?

A. Created the reservation already

The scenario explicitly leaves prior creation possible and unresolved.

3. What is unspecified?

D. Duplicate-handling behavior for the proposed retry

The service's treatment of duplicate requests is not defined in the supplied contract.

D Language checks / page 63

1. Which statement follows from the timeout?

C. The client did not receive a timely reply

A client timeout leaves the service-side result uncertain.

2. Why is a new reservation request risky?

B. The first may already have created a reservation

A repeated new operation may add another effect while the first outcome remains unknown.

3. What does an idempotency key require?

C. Agreed service semantics

A key is useful only when the service contract defines how repeated intent is handled.

4. Which next step is appropriate?

D. Clarify reconciliation and retry behavior, then test it outside production

The contract and controlled evidence are missing, and production changes are not authorized.

07 Conversation answers

Discussing retries and shared state

E Dialogue gaps

Pages 64-65

- 1. uncertain outcome.** Turn 2. Uncertain outcome means the first reservation may exist even though the client did not receive a timely response.
- 2. timeout.** Turn 4. Timeout describes expiration of the client's wait, not a confirmed service-side rejection or absence of a reservation.
- 3. deduplication.** Turn 6. Deduplication behavior is unspecified, so repeated intent cannot be assumed to produce only one reservation.
- 4. idempotency key.** Turn 8. An idempotency key needs a defined service contract for repeated intent rather than merely being added to a request.
- 5. request identity.** Turn 10. Request identity distinguishes a repeated attempt at the original operation from a new reservation with another intended effect.
- 6. status lookup.** Turn 12. Status lookup is a possible reconciliation mechanism that must actually be supported and defined rather than invented.
- 7. Backoff.** Turn 14. Backoff changes timing between attempts but does not itself establish the earlier outcome or prevent duplicate business effects.
- 8. authorization.** Turn 16. Authorization is explicitly absent for production changes, so investigation must not be described as permission to enable the retry.
- 9. shared state.** Turn 18. Shared state is the reservation state whose service-side result may differ from what the timed-out client knows.
- 10. reconciliation.** Turn 20. Reconciliation establishes the actual first-request outcome and supports a defined retry decision rather than assuming failure from silence.

F Transfer conversation

Page 66

1. timeout

Developer: "The client observed a timeout."

The client stopped waiting before it received the service reply.

2. exist

Reviewer: "The first reservation may already exist."

The service may have completed creation before the client timed out.

3. unspecified

Developer: "Duplicate handling remains unspecified."

The service contract supplies no duplicate-handling behavior for this proposal.

4. authorized

Reviewer: "No production change is authorized."

The scenario explicitly withholds permission to enable a live change.

08 Briefing and language checks

Handing over a release with explicit limits

A Briefing checks / page 67

1. Where is R18 deployed?

D. Staging

The scenario establishes a staging deployment only, not production deployment.

2. What is the new search feature's state?

D. Disabled by a feature flag

The search feature is present but disabled by its flag.

3. Which check is unresolved?

A. Compatibility with the previous application version

Compatibility with the prior application has not been tested.

D Language checks / page 72

1. Which status is accurate?

C. R18 is in staging with search disabled

Only the staging deployment and disabled search state are established.

2. Why does optional not settle compatibility?

D. The previous application's behavior with migrated data is untested

Optional describes a field rule, not evidence of the older application's behavior.

3. Which rollback claim is justified?

D. Readiness remains unverified without the relevant compatibility evidence

The earlier application's compatibility with the changed state has not been tested.

4. Who owns the unresolved production decision?

C. Lee

Lee is explicitly identified as the reviewer responsible for the production decision.

08 Conversation answers

Handing over a release with explicit limits

E Dialogue gaps

Pages 73-74

- 1. environment boundary.** Turn 2. Environment boundary limits the deployment claim to staging rather than implying a live production release.
- 2. feature flag.** Turn 4. The feature flag controls whether new search is enabled, which is separate from the code being deployed.
- 3. known limitation.** Turn 6. Known limitation identifies the untested previous-version compatibility rather than presenting optionality as proof.
- 4. Schema compatibility.** Turn 8. Schema compatibility requires relevant evidence about components working with the changed structure, which is not supplied here.
- 5. rollback.** Turn 10. Rollback is a return to an earlier state whose readiness depends on compatibility with the actual migrated data and system state.
- 6. data compatibility.** Turn 12. Data compatibility concerns whether the earlier application can correctly use the actual migrated state, not whether an old executable exists.
- 7. go/no-go decision.** Turn 14. The go/no-go decision is Lee's unresolved production decision, not an automatic consequence of receiving the handoff.
- 8. handoff.** Turn 16. Handoff transfers specific state, evidence gaps, and responsibility so the reviewer does not have to infer readiness.
- 9. release gate.** Turn 18. A release gate is a condition for advancing the release and cannot be treated as satisfied by an unrelated completed deployment.
- 10. readiness claim.** Turn 20. A readiness claim must stay within the completed checks and unresolved decision rather than equating staging deployment with approval.

F Transfer conversation

Page 75

1. staging

Developer: "R18 is deployed in staging."

The only established deployment environment in the scenario is staging.

2. disabled

Reviewer: "New search remains disabled."

The feature flag keeps search disabled despite the completed deployment.

3. untested

Developer: "Prior-version compatibility is untested."

No test of the previous application against the migrated state is established.

4. Lee

Reviewer: "The production decision remains with Lee."

Lee owns the production review and has not made that decision.

Find the move you need

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

01 Clarifying interface contracts

Name the boundary: I want to confirm the profile interface contract.

Identify the proposal: The proposed client sends null.

Avoid borrowing a rule: I would not infer this from another API.

Full phrase set: pages 7-8 | Dialogue: pages 10-11

02 Debugging from reproducible evidence

Report the failing input: The three-row file fails during import.

Report the comparison: The two-row file succeeds.

Propose a narrow change: Change only the suspect date condition.

Full phrase set: pages 16-17 | Dialogue: pages 19-20

03 Giving actionable code-review feedback

Acknowledge the acceptable part: The helper name is clear.

Separate preference and defect: My concern is behavior, not the naming choice.

Request evidence: Add a test for a failed export.

Full phrase set: pages 25-26 | Dialogue: pages 28-29

04 Requesting dependency evidence

Identify the direct dependency: The application uses ParcelKit directly.

Request the actual selection: Which version does the lockfile record?

Ask about usage: Do we have any affected call sites?

Full phrase set: pages 34-35 | Dialogue: pages 37-38

Keep the conversation productive

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

05 Explaining what a test actually exercises

Describe the current setup: The storage double always succeeds.

Identify the gap: The current test never triggers rejection.

Separate call and result: A storage call alone does not prove the right user message.

Full phrase set: pages 43-44 | Dialogue: pages 46-47

06 Comparing performance tradeoffs fairly

Name the conditions: Both runs use the same input and machine.

Report the absolute gain: B lowers the median by fifteen milliseconds.

Calculate the overrun: B exceeds that budget by forty megabytes.

Full phrase set: pages 52-53 | Dialogue: pages 55-56

07 Discussing retries and shared state

State the observation: The client timed out before receiving a reply.

Identify the proposal: The retry would submit a new reservation request.

Clarify identity: Would this carry the same request identity or a new one?

Full phrase set: pages 61-62 | Dialogue: pages 64-65

08 Handing over a release with explicit limits

Identify the release: This handoff concerns R18.

Separate deployment and use: Deployed does not mean enabled.

Name the decision owner: Lee owns the production review.

Full phrase set: pages 70-71 | Dialogue: pages 73-74

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

absolute difference report the absolute difference	51
acceptance criterion state an acceptance criterion	6
acknowledgment distinguish an acknowledgment	59
act act on the save request	42
adapter test the storage adapter	42
application programming interface document the API	5
approval withhold approval pending correction	24
arrange arrange a rejection case	42
assert assert no false success	42
assertion strengthen the assertion	24
assertion add a negative assertion	41
at-least-once delivery discuss at-least-once delivery	59
at-most-once delivery distinguish at-most-once delivery	60
atomic operation define the atomic operation	60
authorization obtain production authorization	60
backoff configure retry backoff	60
backward compatibility assess backward compatibility	6
backward compatibility verify backward compatibility	69
baseline establish a baseline	14
baseline use A as the baseline	50
behavior change flag the behavior change	23
behavioral verification perform behavioral verification	42
benchmark run a benchmark	50
blocking concern explain the blocking concern	24

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

bottleneck identify the bottleneck	51
branch coverage examine branch coverage	42
breaking change identify a breaking change	33
budget compliance check budget compliance	51
call site identify the call site	23
call site search for affected call sites	32
causal claim qualify a causal claim	15
changelog compare the changelog	33
compatibility verify compatibility	33
completion notification gate the completion notification	23
concurrency account for concurrency	60
confounding variable remove a confounding variable	14
consumer identify the consumer	6
contract test add a contract test	6
control flow trace the control flow	23
controlled comparison run a controlled comparison	14
controlled conditions preserve controlled conditions	51
correlation identifier retain the correlation identifier	60
coverage gap report the coverage gap	42
data compatibility test data compatibility	69
date parsing investigate date parsing	14
deduplication define deduplication behavior	60
default value document the default value	5
dependency tree inspect the dependency tree	32

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

deployment	report the deployment	68
deprecation	investigate a deprecation	32
deserialization	verify deserialization behavior	6
deterministic behavior	check deterministic behavior	15
diff	inspect the diff	23
direct dependency	upgrade a direct dependency	32
disabled state	preserve the disabled state	68
distribution	inspect the latency distribution	51
duplicate request	detect a duplicate request	59
duplicate side effect	prevent a duplicate side effect	59
empty field	isolate the empty field	15
empty string	send an empty string	5
entry point	locate the parsing entry point	32
environment boundary	preserve the environment boundary	69
environment parity	establish environment parity	15
error path	exercise the error path	42
exception	capture the exception	15
exception	configure an exception	41
failure path	cover the failure path	23
failure signature	compare the failure signature	15
false negative	avoid a false negative	42
false success	prevent false success	24
feature flag	verify the feature flag	68
field presence	check field presence	6

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

fixture arrange the fixture	41
forward compatibility check forward compatibility	69
go/no-go decision record the go/no-go decision	69
guard condition restore the guard condition	24
handoff prepare the release handoff	69
happy path cover the happy path	42
headroom calculate memory headroom	50
hypothesis test a hypothesis	14
idempotency specify idempotency semantics	59
idempotency key define an idempotency key	59
input fixture preserve the input fixture	14
instrumentation improve instrumentation	15
integration test add an integration test	41
intent confirm the intent	24
interaction verification separate interaction verification	42
interface contract clarify the interface contract	5
known limitation list known limitations	69
latency measure latency	50
lockfile inspect the lockfile	32
log context preserve log context	15
major version interpret the major version	33
malformed input identify malformed input	15
measurement noise account for measurement noise	51
median report median latency	50

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

memory footprint compare the memory footprint	50
migration review the migration	68
migration guide consult the migration guide	32
minimal example reduce to a minimal example	14
minor version check minor-version guarantees	33
mock inspect mock interactions	41
negative assertion verify a negative assertion	41
nonblocking suggestion label a nonblocking suggestion	24
nonproduction environment use a nonproduction environment	15
null value define the null value	5
nullable field declare a nullable field	5
observable behavior preserve observable behavior	24
omitted field distinguish an omitted field	5
optional field omit an optional field	5
optional field add an optional field	68
outlier investigate an outlier	51
overflow quantify the budget overflow	50
p95 distinguish p95 from the median	50
parser inspect the parser	14
partial update apply a partial update	5
patch revision request a patch revision	24
patch version verify patch-version policy	33
percentile report a latency percentile	50
pin review the version pin	33

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

previous version	test the previous version	68
producer	align producer expectations	6
production	authorize production release	68
production behavior	avoid assuming production behavior	51
public interface	compare the public interface	33
pull request	review the pull request	23
race condition	investigate a race condition	60
readiness claim	qualify the readiness claim	69
reconciliation	request outcome reconciliation	59
refactor	separate a refactor from a behavior change	23
regression	investigate a regression	15
regression test	add a regression test	24
rejection	simulate a rejection	41
relative reduction	calculate the relative reduction	51
release candidate	review the release candidate	68
release gate	satisfy the release gate	69
release note	read the release note	32
release owner	name the release owner	69
removal	verify removal separately	32
rename	approve the rename	23
replacement interface	assess the replacement interface	33
reproducible example	share a reproducible example	14
reproduction	obtain a reproduction	14
request identity	preserve request identity	59

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

request payload inspect the request payload	5
resolution reproduce dependency resolution	33
resolved version record the resolved version	32
resource budget respect the resource budget	50
retry define retry behavior	59
retry budget set a retry budget	60
review comment write a review comment	23
review scope clarify review scope	24
rollback assess rollback readiness	68
rollout plan the rollout	68
root cause isolate the root cause	14
row count hold row count constant	15
sample document the sample	51
sandbox test in a sandbox	60
schema revise the schema	6
schema compatibility verify schema compatibility	68
semantic versioning confirm semantic versioning policy	33
sentinel value define a sentinel value	6
serialization inspect serialization behavior	6
shared state reconcile shared state	59
side effect examine the side effect	23
sign-off obtain release sign-off	69
stack trace inspect the stack trace	14
staging deploy to staging	68

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

status lookup define a status lookup	60
stored state preserve stored state	5
stub use a success stub	41
success branch keep the call in the success branch	23
success response return a success response	41
tail latency measure tail latency	50
test double configure the test double	41
test isolation preserve test isolation	41
test oracle define the test oracle	42
throughput measure throughput separately	50
timeout observe a timeout	59
tradeoff explain the tradeoff	51
transaction boundary clarify the transaction boundary	60
transitive dependency inspect a transitive dependency	32
uncertain outcome report an uncertain outcome	59
undefined behavior resolve undefined behavior	6
unit test extend the unit test	41
upgrade evidence assemble upgrade evidence	33
validation rule agree on a validation rule	5
verification evidence provide verification evidence	24
verification matrix update the verification matrix	69
version constraint check a version constraint	32
wire format inspect the wire format	6
workload describe the workload	51

Original cases. Clear boundaries.

These are original fictional teaching conversations, not transcripts of real workers. Names, figures, organizations, and incidents are invented. Sources support terminology and communication aims, not the fictional results or an endorsement of this book.

[1] Google Engineering Practices. What to Look for in a Code Review.

[Open the original source](#)

Context for reviewing functionality, tests, naming, and changed behavior. All pull requests and conversations here are original. Checked 1 October 2026.

[2] Amazon Builders Library. Making Retries Safe with Idempotent APIs.

[Open the original source](#)

Background on uncertain outcomes and retry contracts. The fictional reservation interface is not an AWS service or implementation guide. Checked 1 October 2026.

[3] Python Documentation. unittest.mock.

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Reference for configured test doubles, return values, and raised exceptions. Exercises distinguish simulated behavior from actual dependency coverage. Checked 1 October 2026.

[4] Semantic Versioning 2.0.0.

[Open the original source](#)

Terminology for public interfaces, compatibility, and deprecation. The fictional packages are not assumed to follow this versioning convention. Checked 1 October 2026.

Phil Stilwell

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